

25th European Conference on Computational Biology (ECCB)

Tutorial session T13: 13:45 -17:30

Title: «Democratizing multi-omics data analysis with BiomiX and Nextflow: from single-omics to advanced integration»

Presenters :

Cristian IPERI PhD

Álvaro Fernández-Ochoa

**USZ: Center for Experimental
Rheumatology, Zurich
Switzerland**

**University of Granada,
Spain**

USZ Universitäts
Spital Zürich



PRECISESADS

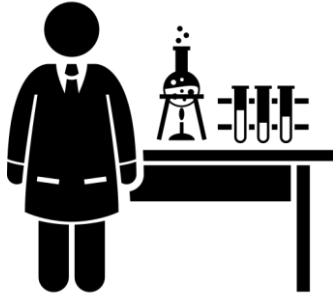
Molecular Reclassification to Find
Clinically Useful Biomarkers for
Systemic Autoimmune Diseases



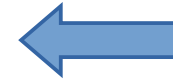
BIOMIX

MULTIOMICS DATA INTEGRATION

03-09-2026



**Access to data and
methods**



Clinicians, Specialists

Bioinformaticians



Deep knowledge of the disease



Extract information from data



Deep knowledge of the disease



Extract information from data



You have a biological question, but you are not a programmer



SEQUIN

Galaxy
EUROPE



**User-friendly tool for single omics
analysis existing but Fragmented**



**Multiomics integration tools with
accessible user interface**



You have a biological question, but you are not a programmer



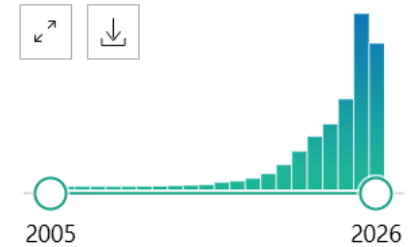
SEQUIN

Galaxy
EUROPE



**User-friendly tool for single omics
analysis existing but Fragmented**

RESULTS BY YEAR



Multiomics research on pubmed
12,031 articles on 24 June 2026.

**Multiomics integration tools with
accessible user interface**

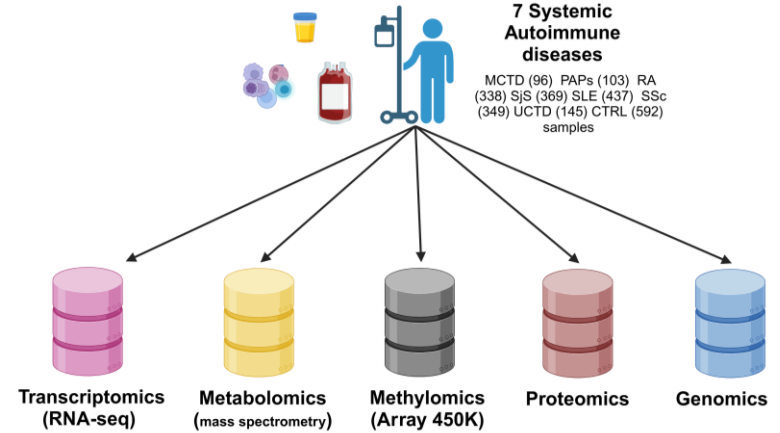


PRECISESADS

Molecular Reclassification to Find
Clinically Useful Biomarkers for
Systemic Autoimmune Diseases

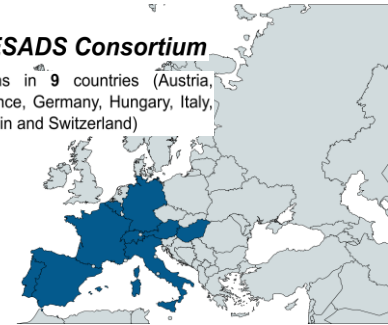
Rationale: Autoimmune disease clinical classifications show high heterogeneity within the disease.

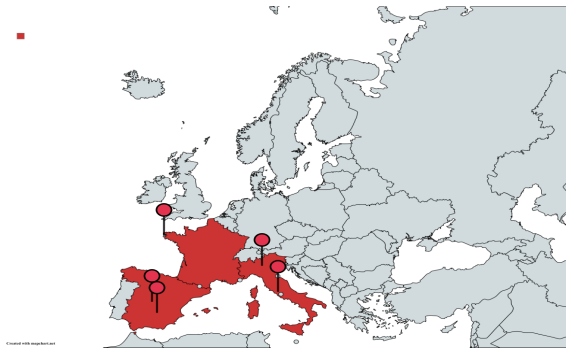
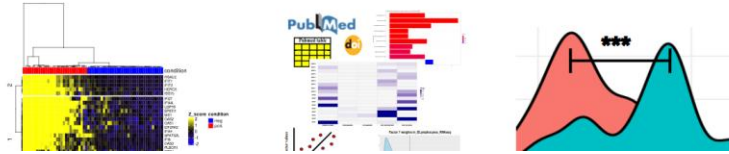
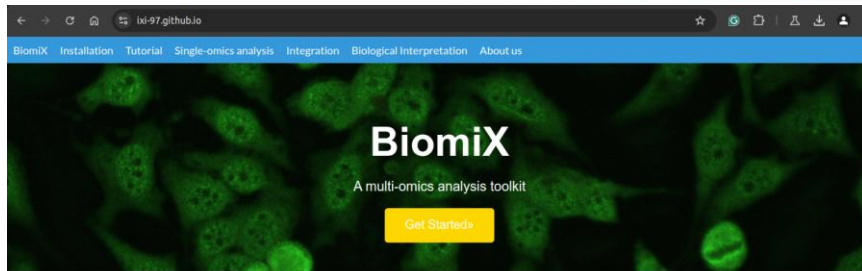
Objective: Reclassification of the autoimmune diseases by biological and molecular criteria



PRECISESADS Consortium

19 institutions in 9 countries (Austria, Belgium, France, Germany, Hungary, Italy, Portugal, Spain and Switzerland)





Software | [Open access](#) | Published: 10 January 2025

BiomiX, a user-friendly bioinformatic tool for democratized analysis and integration of multiomics data

[Cristian Iperi](#), [Álvaro Fernández-Ochoa](#), [Guillermo Barturen](#), [Jacques-Olivier Pers](#), [Nathan Foulquier](#), [Eleonore Bettacchioli](#), [Marta Alarcón-Riquelme](#), [PRECISESADS Flow Cytometry Study Group](#), [PRECISESADS Clinical Consortium](#), [Divi Cornec](#), [Anne Bordron](#) & [Christophe Jamin](#)

[BMC Bioinformatics](#) **26**, Article number: 8 (2025) | [Cite this article](#)

2834 Accesses | 1 Citations | 2 Altmetric | [Metrics](#)



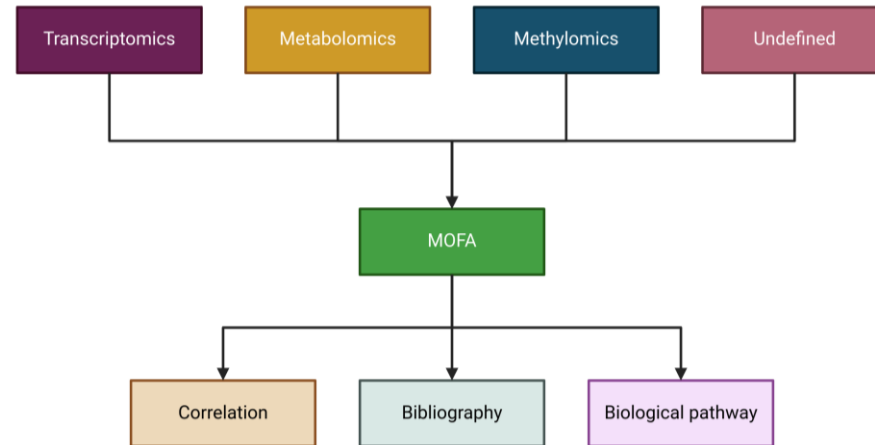
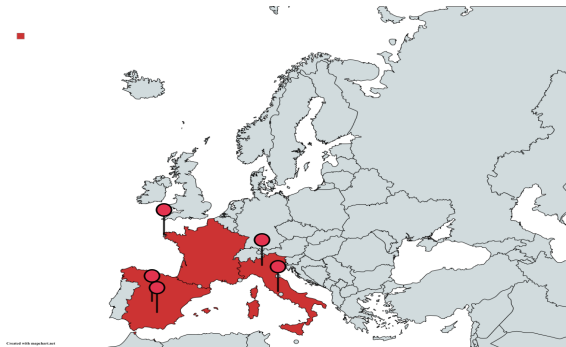
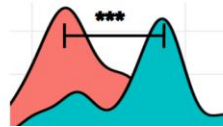
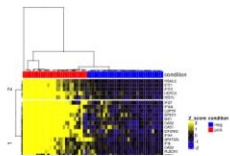
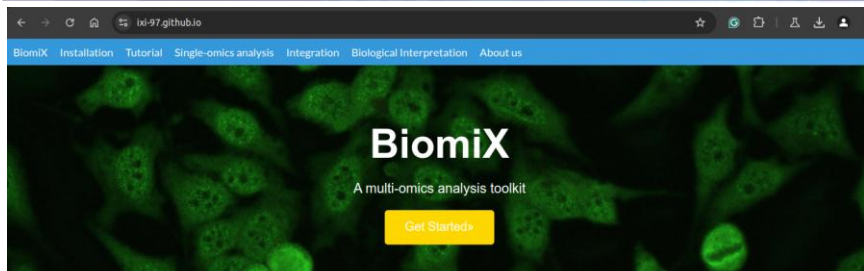
Website: <https://ixi-97.github.io/>



<https://github.com/lxl-97/BiomiX>



<https://ixi-97.github.io/Tutorial.html>



Website: <https://ixi-97.github.io/>



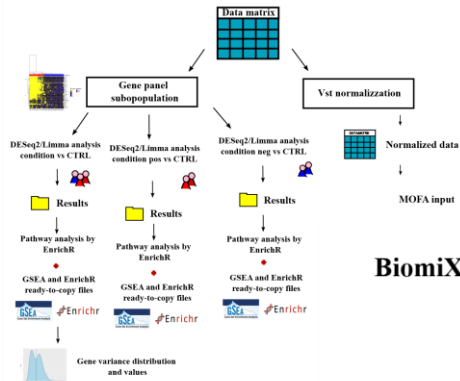
<https://github.com/lxl-97/BiomiX>



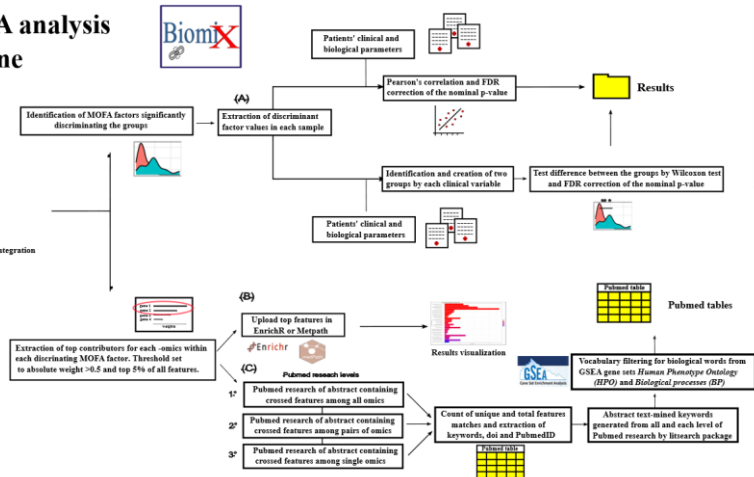
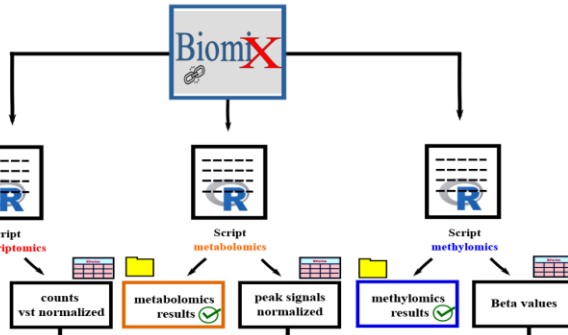
<https://ixi-97.github.io/Tutorial.html>



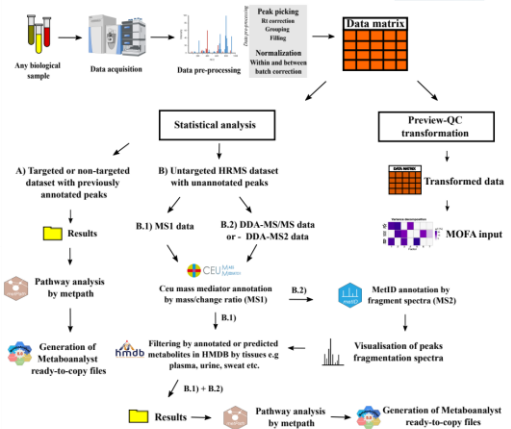
Biomix transcriptomics pipeline



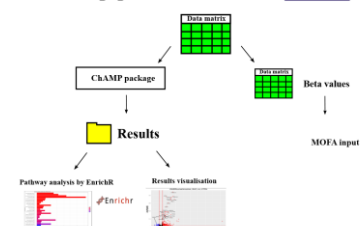
Biomix MOFA analysis pipeline



Biomix metabolomics pipeline



Biomix methylomics pipeline





BiomiX, since then?

Hub of multiomics methods → **accessible to everyone**

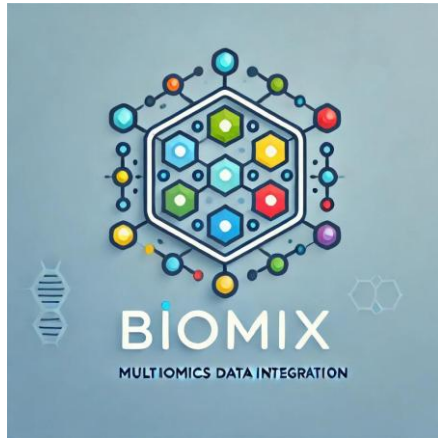
Increase the community of developers and users

Look at the **single and spatial omics** revolution



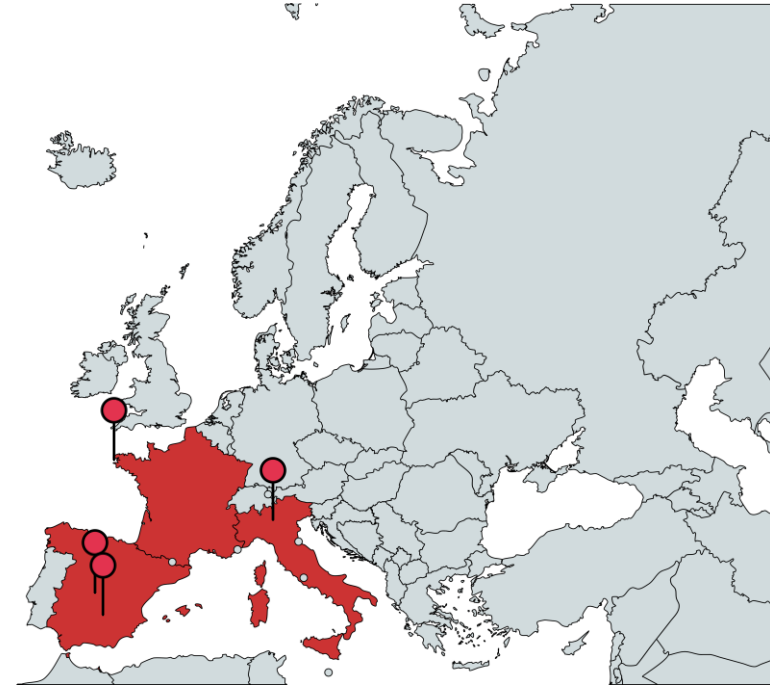


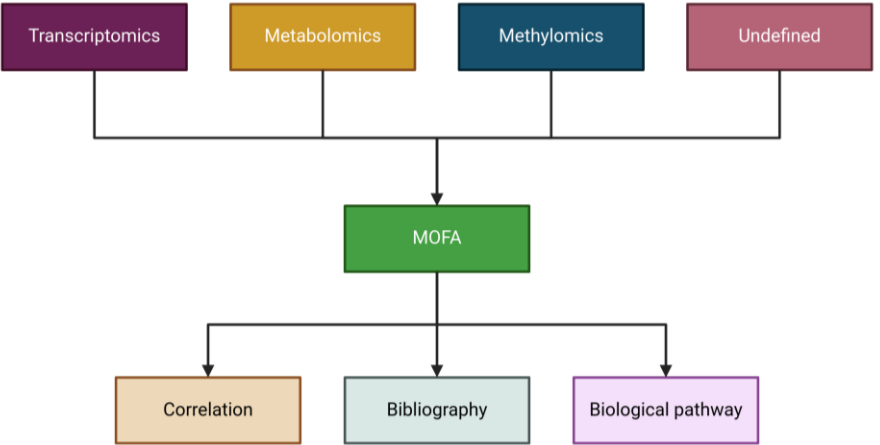
BiomiX, since then?



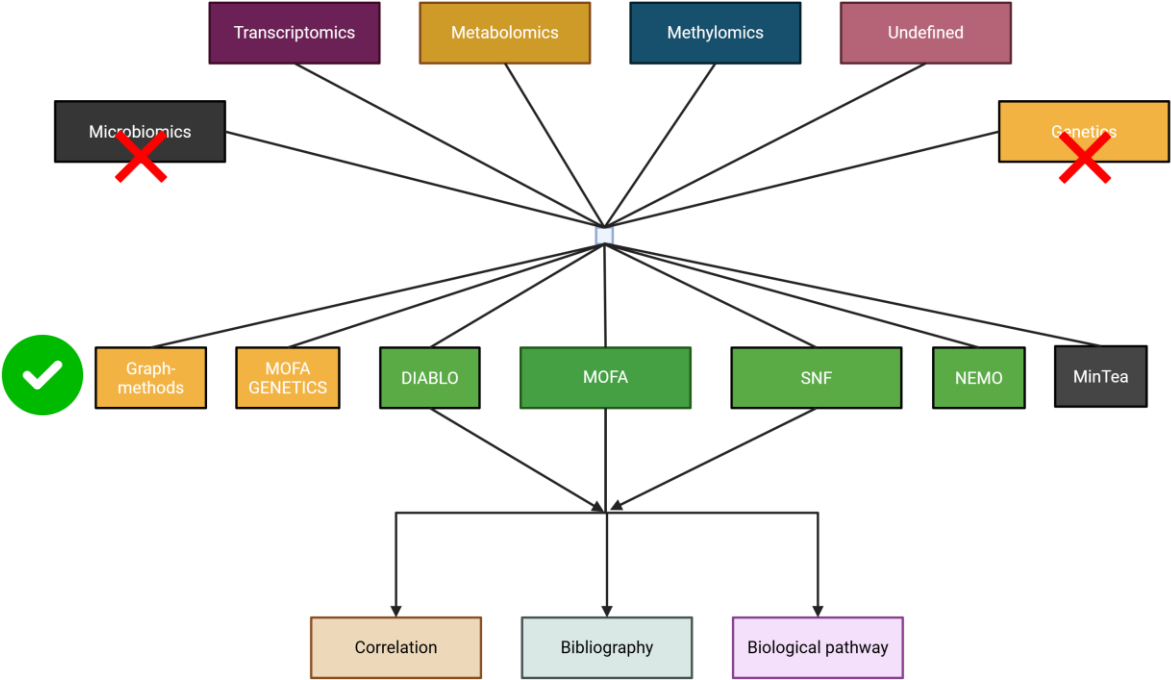
UNIVERSIDAD DE GRANADA

BiomiX consortium





BiomiX V2 2024



BiomiX 2026: Reality (V3.0)



Set up installation



BiomiX

3.0

Home

Installation

Tutorial

Single-omics

Integration

INSTALLATION

Get BiomiX 3.0 running

BiomiX 3.0 uses Docker to ensure a stable, reproducible environment across all platforms.

The launcher scripts handle everything automatically, no command-line knowledge required.

0

Install Docker

BiomiX 3.0 requires Docker to be installed on your system. We recommend [Docker Engine](#) on Linux/Ubuntu, which is the environment BiomiX 3.0 has been built and tested for. New to Docker? Follow our [Docker installation guide](#).

A local installation without Docker is still possible using the legacy install scripts. See the [legacy installation section](#) at the bottom of this page for details.

Slide to be updated with the final version*

1) Open BiomiX website:
<https://biomix-consortium.github.io/>

2) If not done yet install the Docker system for your system (Windows, MacOS, Linux)



Set up installation

The screenshot shows the BiomiX website's installation page. At the top, there is a navigation bar with links: Home, Installation (highlighted in blue), Tutorial, Single-omics, Integration, Parameters, About, and a GitHub button. Below the navigation bar, the main content area has a light blue background. It starts with a section titled '1 Download the BiomiX launcher' with a blue circular icon containing the number 1. Below this title, there is a paragraph: 'Download the launcher script for your operating system from the BiomiX GitHub repository. Place it in a folder of your choice, it will automatically create a `biomix_shared` folder nearby for your input data and results.' Below the paragraph, there are three white buttons with rounded corners, each representing a different operating system: 'BiomiX_Windows.zip' (Windows icon) with the subtext 'Windows, contains .bat and .ps1', 'BiomiX_Start.sh' (Linux icon) with the subtext 'Linux, run from terminal', and 'BiomiX-Launcher-1.0.0.dmg' (MacOS icon) with the subtext 'macOS, open and install'. At the bottom of this section, there is a light blue rounded rectangle containing a small icon and the text: 'All launchers are also available in the BiomiX GitHub repository under the `launchers/` folder.'

Slide to be updated with the final version*

1) Open BiomiX website:
<https://biomix-consortium.github.io/>

2) If not done yet install the Docker system for your system (Windows, MacOS, Linux)

3) Download the BiomiX launcher



Set up installation

Slide to be updated with the final version*

2 Launch BiomiX

The launcher will automatically pull all required Docker modules on first run and open the BiomiX interface in your browser. Select your platform below.

Windows

Linux

macOS

1. Make sure **Docker Desktop is open and running** (look for the whale icon in the system tray).
2. **Double-click** `BiomiX_Start.bat`, a setup window will appear.
3. Choose the **folder** where your input data is located and where results will be saved.
4. Optionally, enter your **NCBI API key** to enhance PubMed search in the biological interpretation module. This is not required, BiomiX works fully without it.
5. Click **Start**. BiomiX will open automatically in your browser at `http://localhost:3838`.

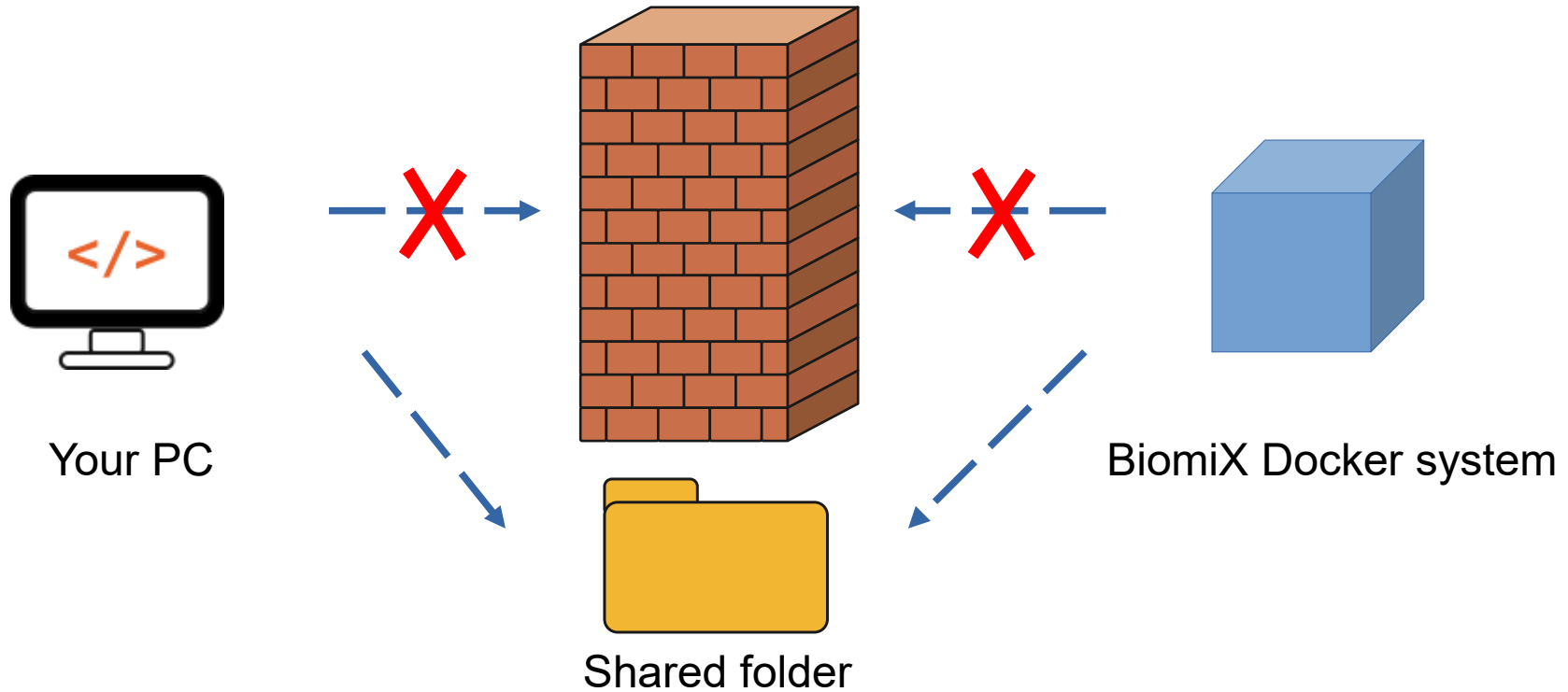


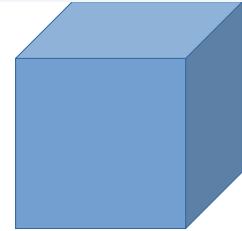
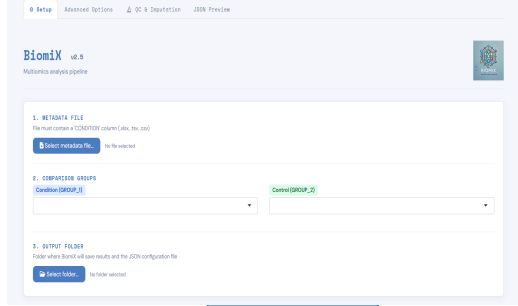
Keep the terminal window open while using BiomiX. Closing it will stop the application.

4) Launch BiomiX

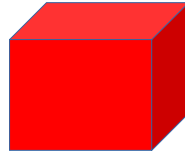


How BiomiX works?

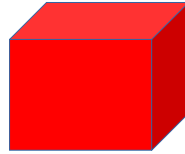




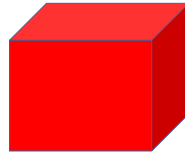
Biomix GUI
Docker



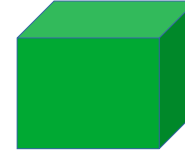
Docker transcriptomics



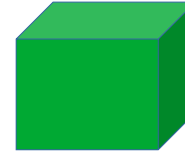
Docker metabolomics



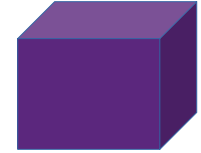
Docker methylomics



Docker MOFA/DIABLO



Docker SNF/NEMO



Docker Interpretation
• Pubmed/Clinical/Enrichment

Single omics dockers

Integration dockers

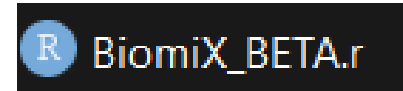
Interpretation dockers



How BiomiX works?

BiomiX interface

→ Json.file →



Master script coordinating
all the analysis.



Datasets of this tutorial

> [J Clin Invest](#). 2018 Jan 2;128(1):427-445. doi: 10.1172/JCI93801. Epub 2017 Dec 11.

Drug-perturbation-based stratification of blood cancer

Sascha Dietrich^{1 2 3 4}, Malgorzata Oles¹, Junyan Lu¹, Leopold Sellner^{2 3}, Simon Anders¹, Britta Velten¹, Bian Wu³, Jennifer Hüllein^{1 3}, Michelle da Silva Liberio³, Tatjana Walther³, Lena Wagner³, Sophie Rabe^{1 2 3}, Sonja Ghidelli-Disse⁵, Marcus Bantscheff⁵, Andrzej K Oleś¹, Mikolaj Slabicki³, Andreas Mock¹, Christopher C Oakes^{6 7}, Shihui Wang³, Sina Oppermann³, Marina Lukas³, Vladislav Kim¹, Martin Sill⁸, Axel Benner⁸, Anna Jauch⁹, Lesley Ann Sutton¹⁰, Emma Young¹⁰, Richard Rosenquist¹⁰, Xiyang Liu³, Alexander Jethwa³, Kwang Seok Lee³, Joe Lewis¹¹, Kerstin Putzker¹¹, Christoph Lutz², Davide Rossi¹², Andriy Mokhir¹³, Thomas Oellerich^{14 15}, Katja Zirlik^{15 16}, Marco Herling¹⁷, Florence Nguyen-Khac¹⁸, Christoph Plass^{7 15}, Emma Andersson¹⁹, Satu Mustjoki¹⁹, Christof von Kalle^{3 15 20}, Anthony D Ho², Manfred Hensel²¹, Jan Dürig^{15 22}, Ingo Ringshausen²³, Marc Zapatka²⁴, Wolfgang Huber^{1 4}, Thorsten Zenz^{2 3 15 25}

Affiliations + expand

PMID: 29227286 PMCID: [PMC5749541](#) DOI: [10.1172/JCI93801](#)

 98 CLL IGHV mutated  74 CLL IGHV unmutated  12 CLL IGHV unknown

1) Chronic lymphocytic leukemia (CLL): EGAS00001001746

> [Microbiol Spectr](#). 2023 Aug 17;11(4):e0057723. doi: 10.1128/spectrum.00577-23. Epub 2023 Jul 31.

Integration of Metabolomics and Transcriptomics Reveals Major Metabolic Pathways and Potential Biomarkers Involved in Pulmonary Tuberculosis and Pulmonary Tuberculosis-Complicated Diabetes

Yunguang Wang¹, Xinxin He², Danna Zheng³, Qiang He⁴, Lifang Sun^{5 6}, Juan Jin^{2 4}

Affiliations + expand

PMID: 37522815 PMCID: [PMC10434036](#) DOI: [10.1128/spectrum.00577-23](#)

 13 HC  13 PTB  13 PTB and DM

2) Pulmonary tuberculosis and diabetes complications: MTBLS7623



Data generation

Data
alignment/QC

Gene count

DGE analysis

Enrichment
analysis



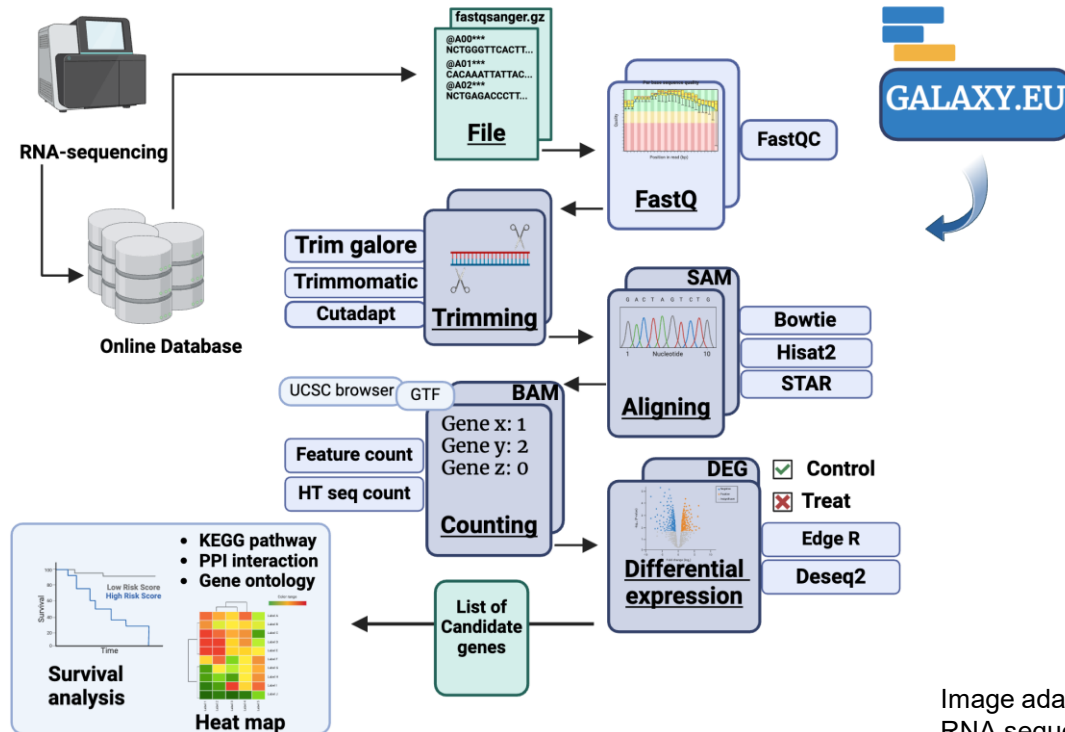
Not Supported by BiomiX
Cluster/High performance
computer running



BiomiX Supported!
Locally running



Transcriptomics analysis CLL

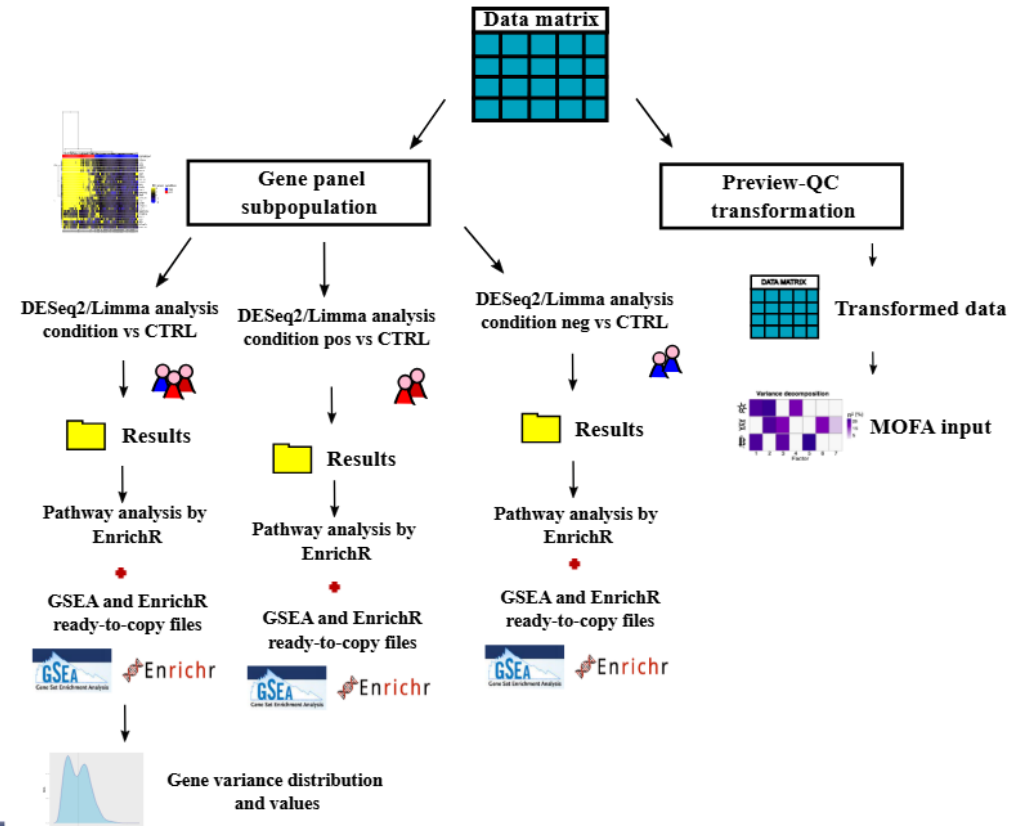
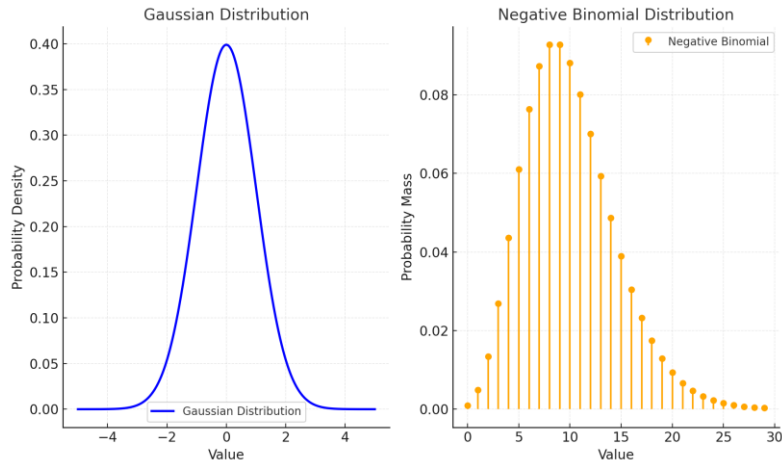
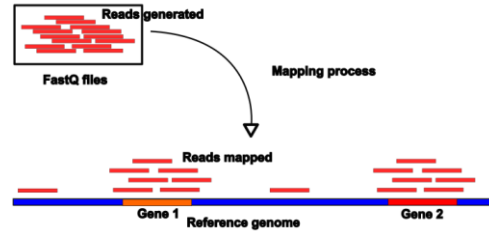


<https://usegalaxy.eu/>

Image adapted from: Sabbaghian A. (2024). User friendly procedure for conducting bulk RNA sequencing analysis using the GALAXY.eu platform. Biorender



BiomiX transcriptomics pipeline





Transcriptomics tutorial



<https://sagarsajeev.medium.com/business-logic-vulnerability-via-idor-6d510f1caea9>



Transcriptomics analysis CLL

1) Open

BiomiX_Start.bat (Windows) **or**

BiomiX_Start.sh (Linux) **or**

BiomiX-Launcher-1.0.0.dmg (MacOS)

- ➔ Create a folder in Desktop called “**shared**” and set it as BiomiX reference directory.

The screenshot shows the BiomiX application window with the following fields and buttons:

- Data folder (your input files and analysis results):** A text box containing the path `C:\Users\crist\Desktop\Shared` and a **Browse...** button to the right.
- NCBI API key (optional - speeds up PubMed searches):** An empty text box.
- Below the API key field, a link: [Get one for free at ncbi.nlm.nih.gov/account/settings](https://ncbi.nlm.nih.gov/account/settings)
- A note at the bottom: *The first run will take longer while Docker downloads BiomiX components.*
- At the bottom right, two buttons: **Cancel** and **Start BiomiX**.



Transcriptomics analysis CLL

1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.

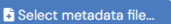
Setup Advanced Options QC & Imputation JSON Preview

Biomix v2.5
Multiomics analysis pipeline



1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

 No file selected



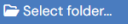
2. COMPARISON GROUPS

Condition (GROUP_1)

Control (GROUP_2)

3. OUTPUT FOLDER

Folder where Biomix will save results and the JSON configuration file

 No folder selected



Transcriptomics analysis CLL

1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.

The screenshot shows the BiomiX v2.5 Multiomics analysis pipeline interface. The 'Setup' tab is active, displaying three sections: '1. METADATA FILE', '2. COMPARISON GROUPS', and '3. OUTPUT FOLDER'. Each section has a 'Select...' button. A file selection dialog is open over the '1. METADATA FILE' section, titled 'Select the metadata file'. The dialog shows a list of files and folders in the 'Data' directory. A red arrow points to the 'Metadata' folder in the list.

name	size	modified	created
Clinical_data		mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
COMMANDS.tsv	309 B	mercoledì 20 mag 2026, 08:37	mercoledì 4 giu 2025, 14:38
COMMANDS_ADVANCED.tsv	2.0 kB	mercoledì 15 apr 2026, 12:15	mercoledì 4 giu 2025, 14:38
COMMANDS_MOFA.tsv	134 B	mercoledì 20 mag 2026, 08:37	mercoledì 4 giu 2025, 14:38
directory.txt	74 B	martedì 9 giu 2026, 11:10	giovedì 5 giu 2025, 11:36
directory_out.txt	32 B	mercoledì 20 mag 2026, 08:36	giovedì 5 giu 2025, 17:40
Example_dataset		mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
Integration		lunedì 8 giu 2026, 09:01	mercoledì 4 giu 2025, 14:38
Metabolomics		martedì 11 nov 2025, 23:34	mercoledì 4 giu 2025, 14:38
Metadata		giovedì 7 mag 2026, 08:47	mercoledì 4 giu 2025, 14:38
Methylomics		martedì 11 nov 2025, 15:34	mercoledì 4 giu 2025, 14:38
modules		martedì 9 giu 2026, 17:06	venerdì 24 apr 2026, 13:26
Package_dependency_map.xlsx	18.7 kB	martedì 5 mag 2026, 08:34	martedì 5 mag 2026, 07:37
Report_parameters		martedì 9 giu 2026, 16:54	mercoledì 4 giu 2025, 14:38
Transcriptomics		martedì 11 nov 2025, 18:04	mercoledì 4 giu 2025, 14:38



Transcriptomics analysis CLL

1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.

Biomix v2.5
Multiomics analysis pipeline

1. METADATA FILE
File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)
Select metadata file... EGAS00001001746_metadata_CLL.xls
184 samples · 3 conditions: mutated, unmutated, unknown

2. COMPARISON GROUPS
Condition (GROUP_1) mutated Control (GROUP_2) unmutated

3. OUTPUT FOLDER
Folder where Biomix will save results and the JSON configuration file
Select folder... No folder selected

Biomix need two metadata columns:
“ID”: Unique sample ID
“CONDITION”: Sample condition





Transcriptomics analysis CLL

1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.

Biomix v2.5
Multiomics analysis pipeline



1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file... EGAS00001001746_metadata_CLL.xls

184 samples · 3 conditions: mutated, unmutated, unknown

2. COMPARISON GROUPS

Condition (GROUP_1)
mutated

Control (GROUP_2)
unmutated

3. OUTPUT FOLDER

Folder where Biomix will save results and the JSON configuration file

Select folder... No folder selected

In docker system it is automatically set up to the shared folder between docker and your pc.



Slot system:
6 slot

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Undefined	☐	Label input 1	☐	Browse file No file selected
Input 2	☐	☐	Undefined	☐	Label input 2	☐	Browse file... No file selected
Input 3	☐	☐	Undefined	☐	Label input 3	☐	Browse file... No file selected
Input 4	☐	☐	Undefined	☐	Label input 4	☐	Browse file... No file selected
Input 5	☐	☐	Undefined	☐	Label input 5	☐	Browse file... No file selected
Input 6	☐	☐	Undefined	☐	Label input 6	☐	Browse file... No file selected

INTEGRATION OPTIONS

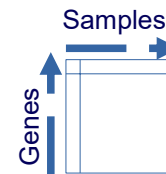
INTEGRATION

☐ Yes ☒ No

Browse file...

EGAS00001001746_transcriptomics.xls

C:/Users/crist/Desktop/BiomiX2.5/Transcriptomics/INPUT/EGAS00001001746_transcriptomics.xls



Upload:
EGAS00001001746_transcriptomics.xls



Normalization and visualisation of the data before the analysis (Shiny app).

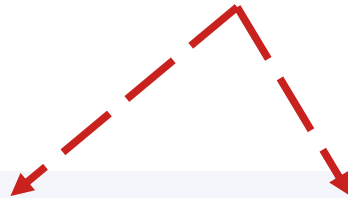
OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Undefined	☐	Label input 1	☐	Browse file... No file selected

Which omics? If not included select undefined (default)



Do you want to perform a single omics analysis or use the slot for the integration?



OMICS INPUTS							
INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Undefined ▾	☐	Label input 1	☐	Browse file... No file selected



Personalized name for your dataset. If “filter” selected will only keep the sample same containing this label.

	Lymphocytes_01	Monocytes_01
Gene 1		
Gene 2		
Gene 3		
Gene 4		
Gene 5		
Gene 6		
Gene 7		

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Undefined ▾	☐	LYMPHOCYTES	☐	Browse file... No file selected



Run your first analysis

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☒	Transcriptomics ▾	☐	RNA	☐	Browse file... EGAS00001001746_transcriptomics.xls C:/Users/crist/Desktop/BioMiX2.5/Transcriptomics/INPUT/EGAS00001001746_transcriptomics.xls
Input 2	☐	☐	Undefined ▾	☐	Label input 2	☐	Browse file... No file selected
Input 3	☐	☐	Undefined ▾	☐	Label input 3	☐	Browse file... No file selected

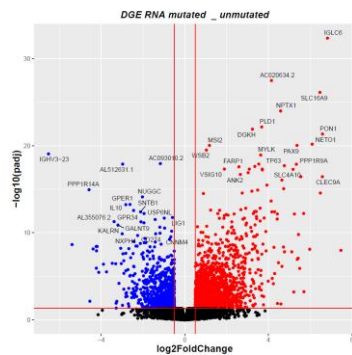
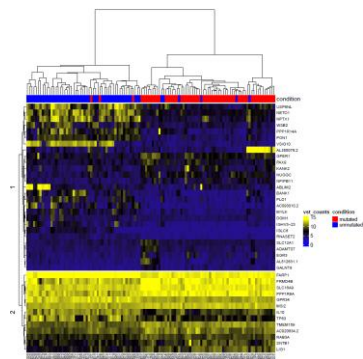
⚙️ Advanced Options

▶ Run BioMiX

📄 Download JSON

🟢 Ready to run

Check in Transcriptomics/Output folder → RNA_mutated_vs_unmutated created!



Input for GSEA program

Pathway_analysis	22/01/2026 11:10	Cartella di file	
Subpopulation_input	22/01/2026 11:09	Cartella di file	
Heatmap_top_genes_RNA_unmutated_vs...	28/04/2026 19:49	Adobe Acrobat D...	41 KB
mutated_unmutated_RNA.tsv	28/04/2026 19:48	File TSV	5.643 KB
mutated_unmutated_RNA_RAW.gct	28/04/2026 19:49	File GCT	58,409 KB
mutated_unmutated_RNA_RAW_meta.tsv	28/04/2026 19:49	File TSV	4 KB
mutated_unmutated_RNA_RAW_meta_gr...	28/04/2026 19:49	File TSV	2 KB
plot_DGE_mutated_unmutated_RNA.pdf	28/04/2026 19:49	Adobe Acrobat D...	1,597 KB
RNA_mutated_unmutated_DOWN_ENRIC...	28/04/2026 19:49	File TSV	8 KB
RNA_mutated_unmutated_GENESDOWN....	28/04/2026 19:49	File TSV	115 KB
RNA_mutated_unmutated_GENESUP.tsv	28/04/2026 19:49	File TSV	274 KB
RNA_mutated_unmutated_UP_ENRICHR.tsv	28/04/2026 19:49	File TSV	19 KB



DGE results all

	baseMean	log2FoldCI	lfcSE	stat	pvalue	padj		
ENSG0000	1906.988	6.828504	0.533117	12.80864	1.4667914	4.56362816482032e-33		
ENSG0000	6.345746	4.146017	0.349858	11.85056	2.1377006	3.32551409152136e-28		
ENSG0000	111.8656	6.465554	0.559783	11.55012	7.3721385	7.64564487909312e-27		
ENSG0000	386.3032	4.586128	0.413405	11.09356	1.3481534	1.04862742655686e-24		

Only negative

Only Positive

Pathway_analysis	22/01/2026 11:10	Cartella di file	
Subpopulation_input	22/01/2026 11:09	Cartella di file	
Heatmap_top_genes_RNA_unmutated_vs...	28/04/2026 19:49	Adobe Acrobat D...	41 KB
mutated_unmutated_RNA.tsv	28/04/2026 19:48	File TSV	5.643 KB
mutated_unmutated_RNA_RAW.gct	28/04/2026 19:49	File GCT	58.409 KB
mutated_unmutated_RNA_RAW_meta.tsv	28/04/2026 19:49	File TSV	4 KB
mutated_unmutated_RNA_RAW_meta_gr...	28/04/2026 19:49	File TSV	2 KB
plot_DGE_mutated_unmutated_RNA.pdf	28/04/2026 19:49	Adobe Acrobat D...	1.597 KB
RNA_mutated_unmutated_DOWN_ENRIC...	28/04/2026 19:49	File TSV	8 KB
RNA_mutated_unmutated_GENESDOWN....	28/04/2026 19:49	File TSV	115 KB
RNA_mutated_unmutated_GENESUP.tsv	28/04/2026 19:49	File TSV	274 KB
RNA_mutated_unmutated_UP_ENRICHR.tsv	28/04/2026 19:49	File TSV	19 KB

Input for GSEA program



GSEA input



Enrichr

Input negative gene EnrichR

Input positive gene EnrichR

Pathway_analysis	22/01/2026 11:10	Cartella di file	
Subpopulation_input	22/01/2026 11:09	Cartella di file	
Heatmap_top_genes_RNA_unmutated_vs...	28/04/2026 19:49	Adobe Acrobat D...	41 KB
mutated_unmutated_RNA.tsv	28/04/2026 19:48	File TSV	5.643 KB
mutated_unmutated_RNA_RAW.gct	28/04/2026 19:49	File GCT	58.409 KB
mutated_unmutated_RNA_RAW_meta.tsv	28/04/2026 19:49	File TSV	4 KB
mutated_unmutated_RNA_RAW_meta_gr...	28/04/2026 19:49	File TSV	2 KB
plot_DGE_mutated_unmutated_RNA.pdf	28/04/2026 19:49	Adobe Acrobat D...	1.597 KB
RNA_mutated_unmutated_DOWN_ENRIC...	28/04/2026 19:49	File TSV	8 KB
RNA_mutated_unmutated_GENESDOWN....	28/04/2026 19:49	File TSV	115 KB
RNA_mutated_unmutated_GENESUP.tsv	28/04/2026 19:49	File TSV	274 KB
RNA_mutated_unmutated_UP_ENRICHR.tsv	28/04/2026 19:49	File TSV	19 KB



Enrichr

[Login](#) | [Register](#)

130,228,543 sets analyzed
473,654 terms
228 libraries

[Analyze](#)[What's new?](#)[Libraries](#)[Gene search](#)[Term search](#)[About](#)[Help](#)

Input data

Expand a gene, a term, or a variant into a gene set:



Try an example

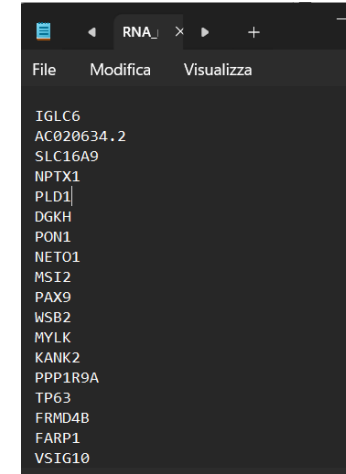
Include the top 100 most relevant genes



Drop a file or paste a set of Entrez gene symbols on each row in the textbox below. You can try a gene set **example**. Also, you can now try adding a **background**.

Drop a file or paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box

0 gene(s) entered



Enrichr

[Login](#) | [Register](#)[Transcription](#)[Pathways](#)[Ontologies](#)[Diseases/Drugs](#)[Cell Types](#)[Misc](#)[Legacy](#)[Crowd](#)

Description No description available (100 genes)

Reactome Pathways 2024

CDC42 GTPase Cycle
RHO GTPase Cycle
VEGFA-VEGFR2 Pathway
RHO GTPase Cycle
Signaling by GTPases

WikiPathways 2024 Human

Non Genomic Actions Of 1 25 Dihydroxyvitamin D3
Glycerolipids And Glycerophospholipids WP
EGF/EGFR Signaling WP437
Endothelin Pathways WP2197
Melatonin Metabolism And Effects WP3298

BioPlanet 2019

GnRH signaling pathway
Delta Np63 pathway
PKC-catalyzed phosphorylation of inhibitory
EGF/EGFR signaling pathway
Phosphatidylinositol signaling system



	A	B	C	D	E
1	#1.2				
2		52141	28		
3	NAME	Description	32140242	32140244	32141679
4	ENSG00000000003	NA	0	1.03138546799503	5.25875398402679
5	ENSG00000000005	NA	0	0	0.39440654880201
6	ENSG000000000419	NA	224.658295044468	207.308479067001	215.608913345099
7	ENSG000000000457	NA	411.72436542014	319.729495078459	337.874943473722
8	ENSG000000000460	NA	148.578792738572	86.6363793115826	101.231014192516
9	ENSG000000000938	NA	2134.7013294066	1614.11825741222	6782.47795089856
10	ENSG000000000971	NA	6.26537077813257	2.06277093599006	2.6293769920134
11	ENSG000000001036	NA	107.406356196558	111.389630543463	157.762619520994
12	ENSG000000001084	NA	247.929672220389	390.895092370117	210.350159361072
13	ENSG000000001167	NA	351.755816543728	257.846366998758	269.511141681373
14	ENSG000000001460	NA	48.3328602884512	45.3809605917813	74.9372442723818
15	ENSG000000001461	NA	508.390085997042	450.715449513828	590.295134707008
16	ENSG000000001497	NA	412.619418388445	414.616958134002	395.721237298016
17	ENSG000000001561	NA	239.874195505647	369.235997542221	437.791269170231
18	ENSG000000001617	NA		0.309415640398509	3.9440654880201
19	ENSG000000001626	NA		0.103138546799503	9.20281947204689
20	ENSG000000001629	NA	848.51021395281	877.709033263771	1020.1982729012
21	ENSG000000001630	NA	332.959704209331	354.796600990291	499.581628482546

Expression matrix in .gct format



Metadata_gsea

GSEA 4.1.0 (Gene set enrichment analysis)

File Downloads Help

Steps in GSEA analysis

- Load data
- Run GSEA
- Leading edge analysis
- Enrichment Map Visualization
- Tools
 - Run GSEAPreranked
 - Collapse Dataset
 - Chip2Chip mapping
- Analysis history

Home Run Gsea x Load data x

Gsea: Set parameters and run enrichment tests

Required fields

Expression dataset

Gene sets database

Number of permutations

Phenotype labels

Collapse/Remap to gene symbols

Permutation type

Chip platform

Basic fields

Advanced fields

Reset Last Command Run

16:31:45 61M of 124M

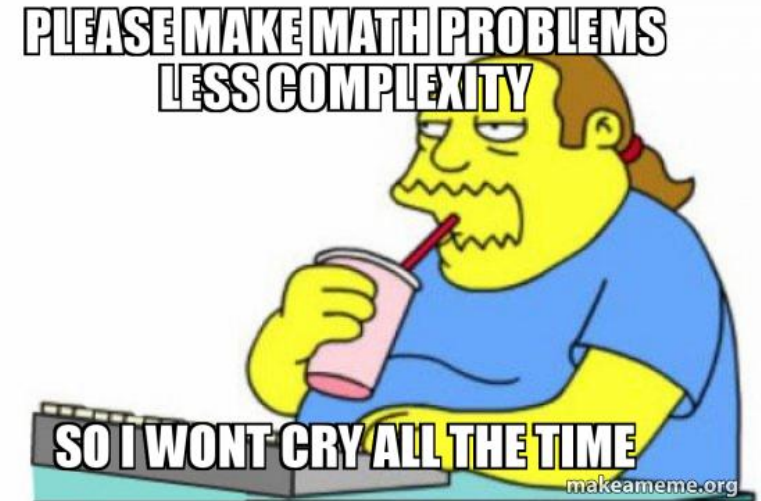


Analysis customization

Just running default parameters can not be sufficient,

What if I want

- Increase the log2FC threshold
- Select only Male patients
- Stratify my patients by IFN signature





Analysis customization

Advanced Options

Run Biomix

Download JSON

- Log2 fold change
- Pvalue
- N* gene in the heatmap
- Sample selection by gene panel

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

TRANSCRIPTOMICS OPTIONS

Log2FC threshold 0,5 P.adj threshold 0,05 Gene panel file (optional) Browse... No file (optional)

METABOLOMICS OPTIONS

Log2FC threshold 0,5 P.adj threshold 0,05

METHYLOMICS OPTIONS

P.adj threshold 0,05 Log2FC threshold 0,15 Array type 450K

GENE PANEL AND POSITIVITY CRITERIA

N° genes Z-score > 2 10 N° genes Z-score > 1 10 N° top DE genes in heatmap 20 Subgrouping ☐ Enable subgrouping

CLUSTERING OPTIONS

Clustering distance euclidean Clustering method ward.D2

COMPUTATIONAL RESOURCES / GENERAL INTEGRATION OPTIONS

CPU threads 3 N° input features for integration 5000



Select gene panel file

	name	size	modified	created
	ALL_SORTED_CELLS_MATRIX_ENSG.tsv	135.6 MB	mercoledì 30 ago 2023, 10:31	mercoledì 3 set 2025, 09:00
	EGAS00001001746_transcriptomics.tsv	23.9 MB	mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
	EGAS00001001746_transcriptomics.xls	56.4 MB	mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
	GENE_PANEL.txt	211 B	mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
	GENE_PANEL.xls	6.7 kB	mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
	MTBLS7623_RNA_seq.tsv	6.6 MB	mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
	MTBLS7623_RNA_seq.xls	18.1 MB	mercoledì 4 giu 2025, 14:38	mercoledì 4 giu 2025, 14:38
	Transcriptomics_screen_RA.tsv	28.4 MB	mercoledì 12 nov 2025, 01:04	giovedì 11 dic 2025, 12:06
	Transcriptomics_screen_RA_Debug.tsv	28.4 MB	mercoledì 12 nov 2025, 01:04	giovedì 11 dic 2025, 12:18
	Transcriptomics_screen_RA_Debug.xlsx	19.6 MB	giovedì 11 dic 2025, 12:20	giovedì 11 dic 2025, 12:20
	Validation_MARKER_vs_GENE_PANEL.tsv	278 B	lunedì 2 feb 2026, 13:52	domenica 1 feb 2026, 12:55

← IFN gene panel

Cancel

Select

3

5000

Setup

Advanced Options

QC & Imputation

JSON Preview

ntTea Metadata Filtering

TRANSCRIPTOMICS OPTIONS

Gene panel file (optional)

Browse...

No file (optional)

METABOLOMICS OPTIONS

METHYLOMICS OPTIONS

Array type

450K

GENE PANEL AND POSITIVITY CRITERIA

N° top DE genes in heatmap

20

Subgrouping

☐ Enable subgrouping

CLUSTERING OPTIONS

Method

GENERAL RESOURCES / GENERAL INTEGRATION OPTIONS



Analysis customization

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diabolo SNF / NEMO MintTea Metadata Filtering

Column choices are populated automatically from the metadata file loaded in the Setup screen. Load the metadata file first if the dropdowns are empty.

SAMPLE FILTERING RULES

	COLUMN	DATA TYPE	THRESHOLD / VALUE	
F1	GENDER	Categorical	== male	☒
F2	ID	ID	e.g. 50 or GroupA	☐
F3	ID	ID	e.g. 50 or GroupA	☐
F4	ID	ID	e.g. 50 or GroupA	☐

1 active filter(s):

- Filter 1: GENDER [Categorical] == male

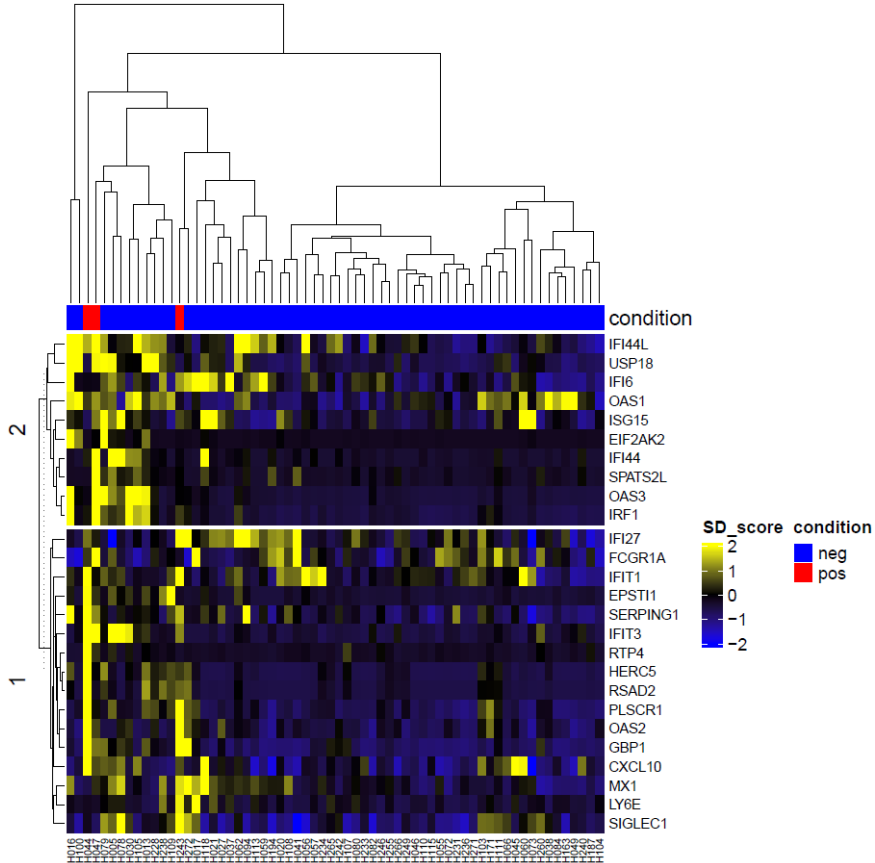
Categorical

== equal to
!= different to

Numerical

> (>=) major of
< (<=) minor of
== equal to

Always space*



Directory:
\\Integration\INPUT\RNA_mutated_vs_unmutated

Conclusion: IFN cannot discriminate CLL



Other functions?

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☒	☒	Undefined	☐	Label input 1	☐	Browse file... No file selected

Normalization and interactive visualisation of the data before the analysis (Shiny app).

⚙️ Advanced Options

Run BiomiX

Download JSON

✅ Ready to run



Same structure, different omics

Advanced Data Normalization and Visualization

Plot Data Table Outliers Download

Select X Variable:
ENSG00000211899

Select Y Variable:
ENSG00000198804

Update Scatter Plot

Transformation methods

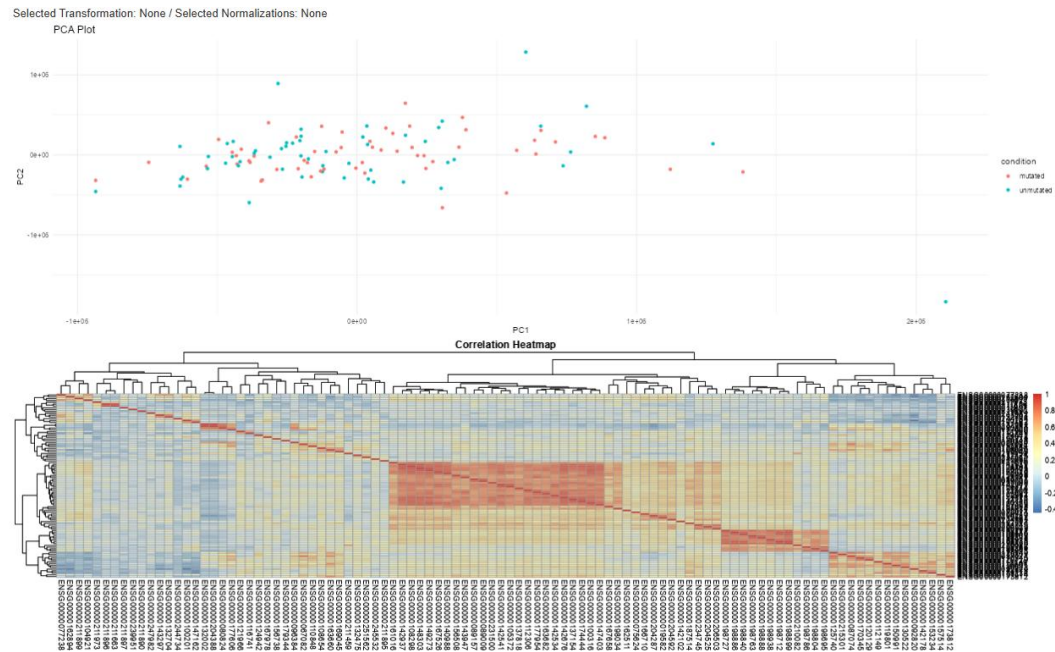
☒ Logarithmic
☐ Median Centralization
☐ Mean Centralization

Normalization methods

☐ Z-Score
☐ MAD
☐ Quantile Normalization
☐ Loess Normalization
☐ VST (Variance Stabilizing Transformation)

Choose Column for Coloring:
condition

Apply Normalization Reset to Original Data Download Current Plot Close App





Same structure, different omics

Advanced Data Normalization and Visualization

[Plot](#) [Data Table](#) [Outliers](#) [Download](#)

Select X Variable:

ENSG00000211899

Select Y Variable:

ENSG00000198804

Update Scatter Plot

Transformation methods

☒ Logarithmic

☐ Median Centralization

☐ Mean Centralization

Normalization methods

☐ Z-Score

☐ MAD

☐ Quantile Normalization

☐ Loess Normalization

☐ VST (Variance Stabilizing Transformation)

Choose Column for Coloring:

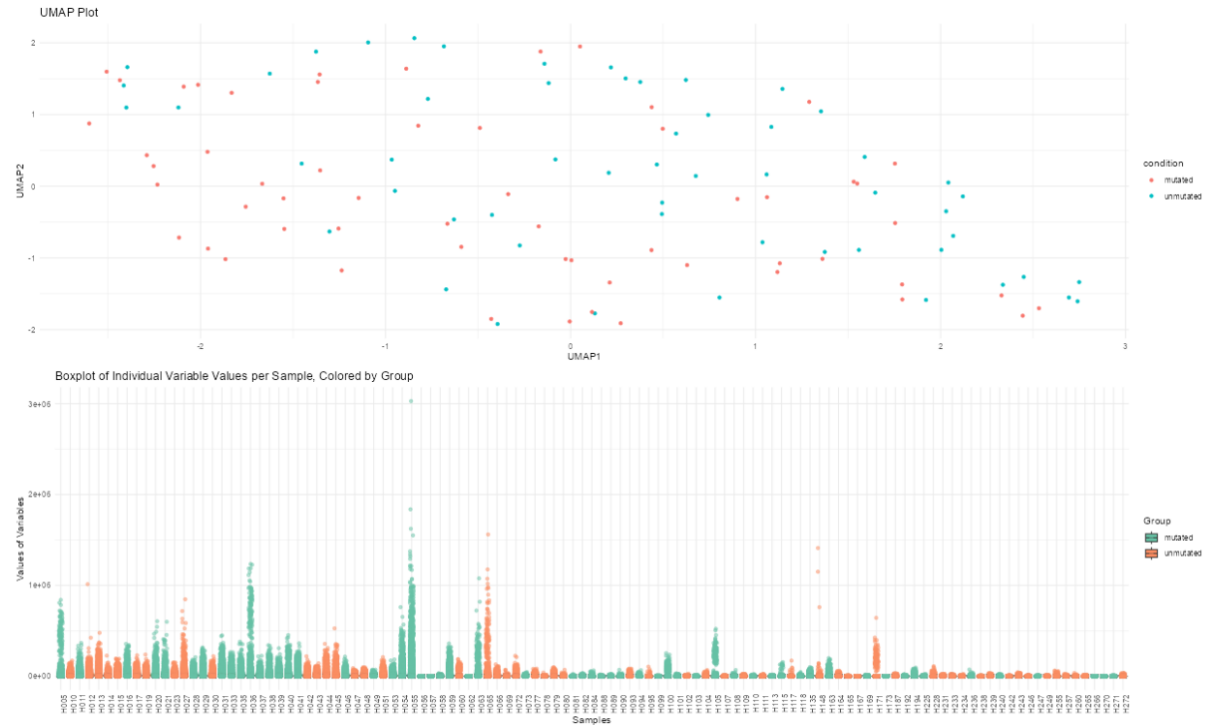
condition

Apply Normalization

Reset to Original Data

Download Current Plot

Close App





Other functions?

Setup

Advanced Options

QC & Imputation

JSON Preview

QC & IMPUTATION

Choose Imputation Method:

Replace with 0

Variable Missingness Threshold (%)

50

Sample Missingness Threshold (%)

50

Impute the matrix

Filter Variables and Samples

Reset to Original Data

Upload Data

Boxplot by Variables

Barplot by Samples

Summary Table

Data Table

Download

Upload a File

Browse... No file selected

☐ Transpose Data

☐ Remove First Row

☐ Remove First Column

Load Data

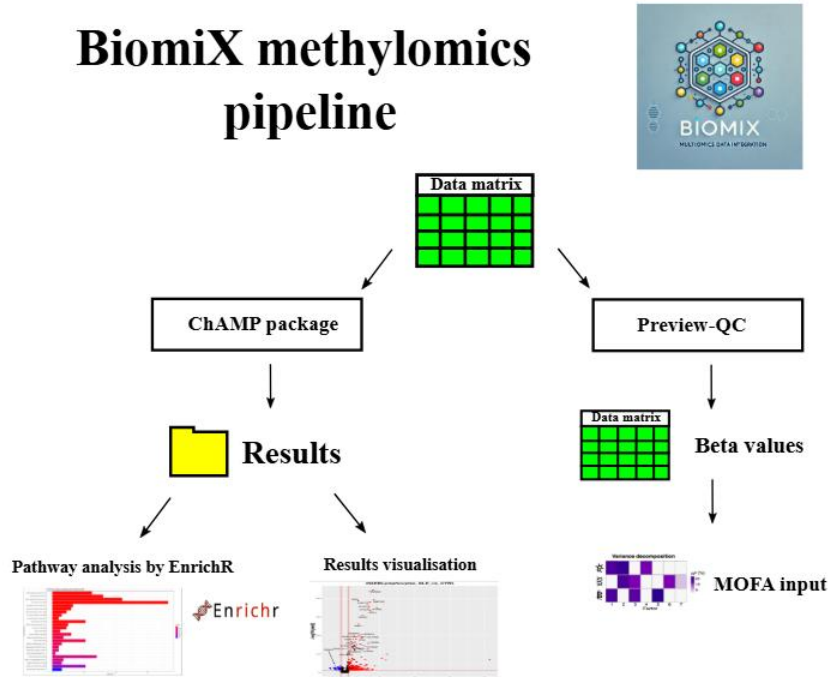
QC & Imputation

- Imputation of missing data
- Removal of sample and variables with NaN data
- Save of the new table



Same structure, different omics

BiomiX methylomics pipeline



1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder. (Metadata slot)

2) Load EGAS00001001746_methylomics.tsv from Methylomics folder. (Input 2)

* To avoid re-running also the transcriptomics analysis deselect “single omics” from the slot 1.

OMICS INPUTS					
INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL
Input 1	☐	☐	Transcriptomics	☐	RNA
Input 2	☐	☒	Methylomics	☐	Methylomics



Metabolomics analysis PTB

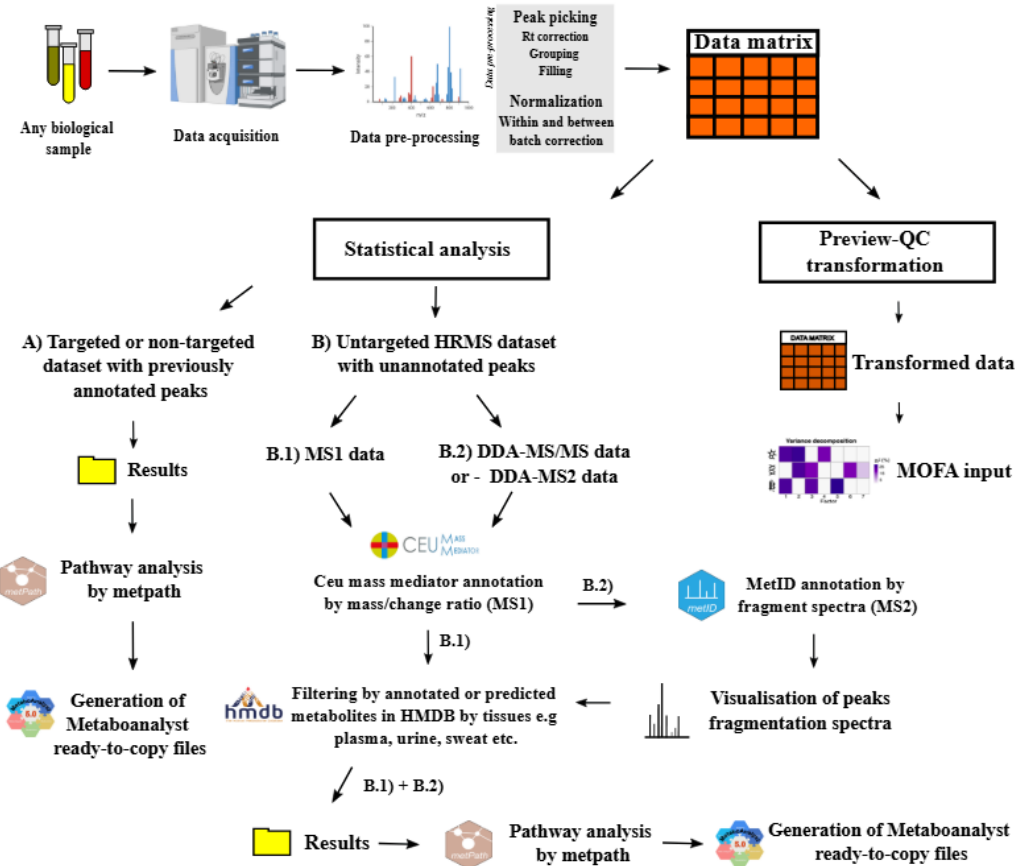
Álvaro Fernández-Ochoa

University of Granada,
Spain



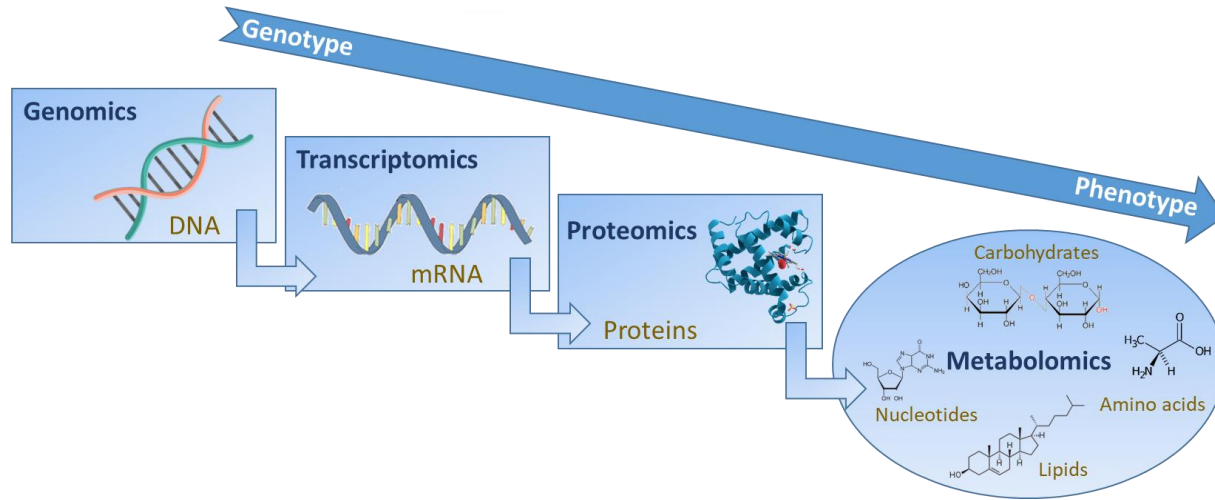
UNIVERSIDAD
DE GRANADA

BiomiX metabolomics pipeline





What is metabolomics?



metabolite levels closely associate with phenotype

metabolomics

analysis of low molecular weight metabolites in biological fluids

"Comprehensive profiling of metabolites key to obtain in depth insight into disease biochemistry"

Molecular weight < 1500 Da



Metabolomics approaches

Targeted metabolomics



Untargeted metabolomics



*BIOMIX supports data
from both approaches*



Predefined

Storage

Previously known metabolites

Identification of signals of interest

Absolute quantification

Relative quantification

Higher sensitivity

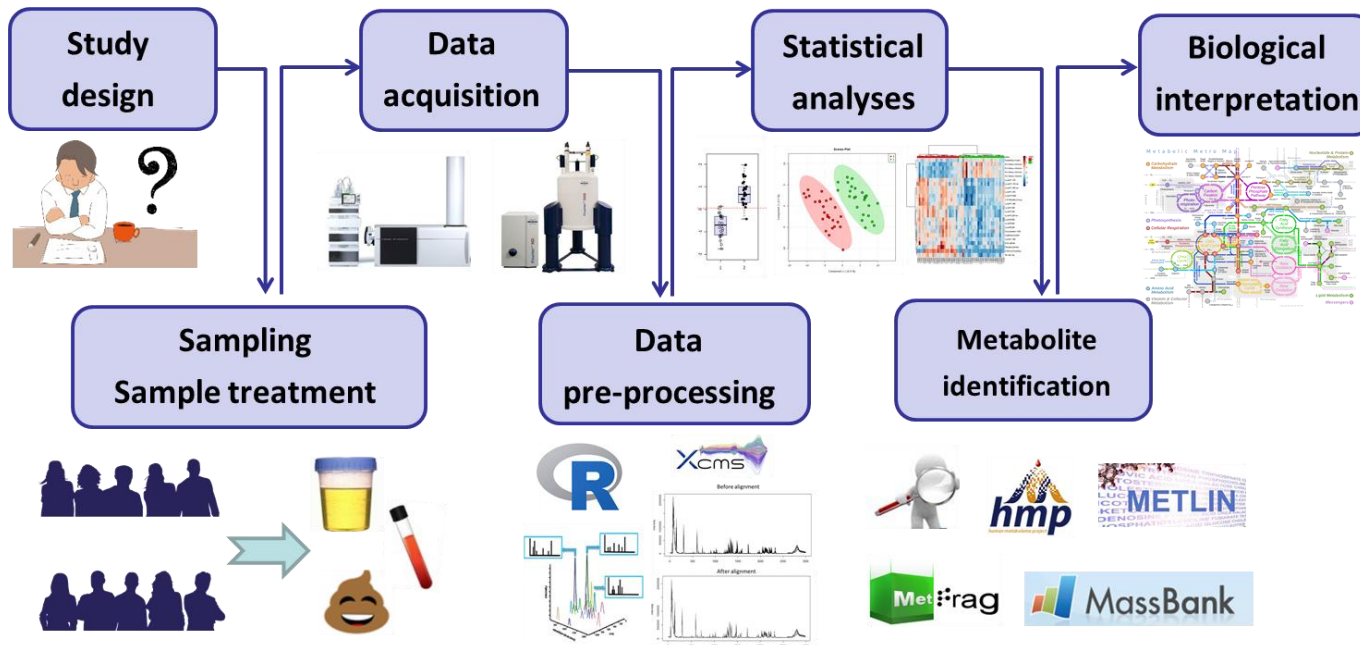
Lower sensitivity

Simplicity in data processing

Complexity in data processing



Metabolomics workflows





COMPUTATIONAL TOOLS

From raw to processed data

Data acquisition

- Data conversion from vendor-specific to open formats (mzML, abf, etc)

Pre-processing

- Mass/peak detection
- Noise filtering
- Peak deconvolution
- Isotope filtering
- Alignment
- Gap-filling
- Duplicate filtering

Normalization

- Drift Correction
- Batch effect correction
- Low-quality feature identification
- Missing value imputation
- Transformation
- Scaling

Statistics

Annotation

Biological interpretation



COMPUTATIONAL TOOLS

From raw to processed data

× Not covered by BiomiX

Data acquisition

- Data conversion from vendor-specific to open formats (mzML, abf, etc)

Pre-processing

- Mass/peak detection
- Noise filtering
- Peak deconvolution
- Isotope filtering
- Alignment
- Gap-filling
- Duplicate filtering

Normalization

- Drift Correction
- Batch effect correction
- Low-quality feature identification
- Missing value imputation
- Transformation
- Scaling

Statistics

Annotation

Biological interpretation

✓ Covered by BiomiX

Sophisticated QC-based methods (drift/batch-effect correction, feature filtering) are NOT covered by BiomiX in the current version

Transformation, scaling, NA imputation & several normalization methods: covered

BiomiX requires an already pre-processed data matrix as input.

Raw data pre-processing and QC-based drift/batch-effect correction must be performed beforehand, following proper QA/QC criteria (especially in untargeted metabolomics).



Data pre-processing

Data processing concepts	Definition
<i>Preprocessing</i>	
Data conversion	Conversion of vendor-specific raw data formats into open formats, such as mzML, mzXML, or ABF, enabling interoperability across analysis software
Peak picking	Detection of ion signals above a defined noise threshold, enabling the detection of potential metabolite signals
Noise filtering	Removal of low-intensity signals that do not represent true molecular features
Peak deconvolution	Separation of overlapping chromatographic peaks corresponding to different compounds that co-elute during chromatographic separation
Alignment	Correction of retention-time shifts across samples to ensure that the same molecular features are matched across datasets
Peak integration	Calculation of peak areas or intensities to estimate metabolite abundance across samples
Grouping of related features	Clustering of isotopes, adducts, in-source fragments, and multimers to represent a single metabolite as one molecular feature



Data pre-processing

Data processing concepts

Definition

Data correction

Quality control assessment



Evaluation of analytical reproducibility and instrument stability using QC samples

Blank subtraction/ contaminant removal

Removal of background signals, contaminants, or solvent-derived peaks

Drift correction

Correction of systematic signal variation caused by instrumental fluctuations over time during a single analytical batch or run sequence, often monitored using QC samples.

Batch normalization

Adjustment of signal intensities to reduce inter-batch variability when samples are analyzed across multiple experimental runs or instruments, ensuring comparability across batches.

Missing value imputation



Estimation or replacement of missing values to allow statistical analysis.

Data transformation



Mathematical transformation of signal intensities (e.g., log, square root, cube root) to stabilize variance, reduce skewness, and make data more normally distributed for statistical analysis.

Scaling



Adjustment of metabolite intensities to standardize variance across features (e.g., autoscaling, Pareto scaling), to give equal importance to low- and high-intensity metabolites in multivariate statistical analysis.



Data pre-processing



Normalization and visualisation of the data before the analysis (Shiny app).

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE
Input 1	☒	☒	Metabolomics

QC & IMPUTATION

Choose Imputation Method:

Replace with 0

Variable Missingness Threshold (%)

50

Sample Missingness Threshold (%)

50

Impute the matrix

Filter Variables and Samples

Reset to Original Data

Advanced Data Normalization and Visualization

Plot Data Table Outliers Download

Select X Variable:

ENSG00000211899

Select Y Variable:

ENSG00000198604

Update Scatter Plot

Transformation methods

☒ Logarithmic

☐ Median Centralization

☐ Mean Centralization

Normalization methods

☐ Z-Score

☐ MAD

☐ Quantile Normalization

☐ Loess Normalization

☐ VST (Variance Stabilizing Transformation)

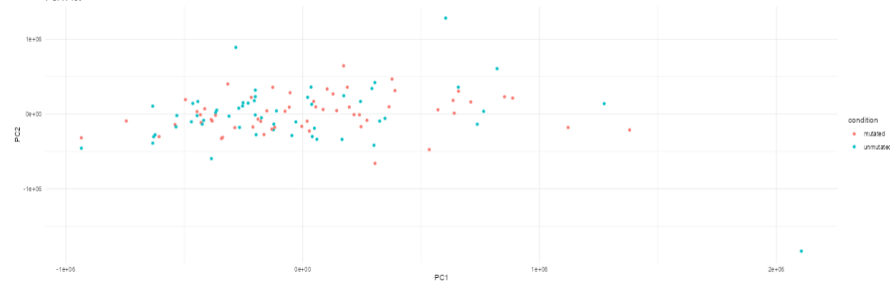
Choose Column for Coloring:

condition

Apply Normalization Reset to Original Data Download Current Plot Close App

Selected Transformation: None / Selected Normalizations: None

PCA Plot





COMPUTATIONAL TOOLS

- Data preprocessing
- Statistical analysis
- Metabolite annotation
- Biological interpretation

Software and specialized platforms



Open-source platforms



Workflow4metabolomics



MS-DIAL



mzmine

Vendor software



Open-source programming languages

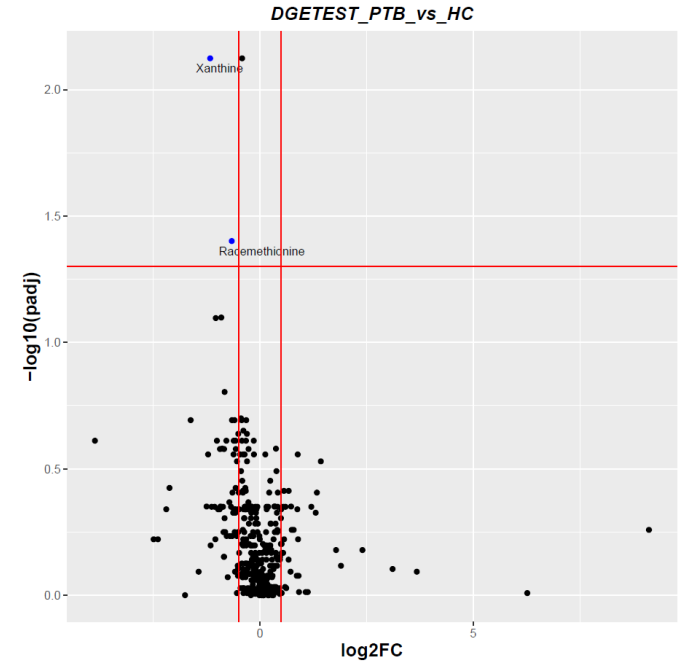




Biomix — Statistics

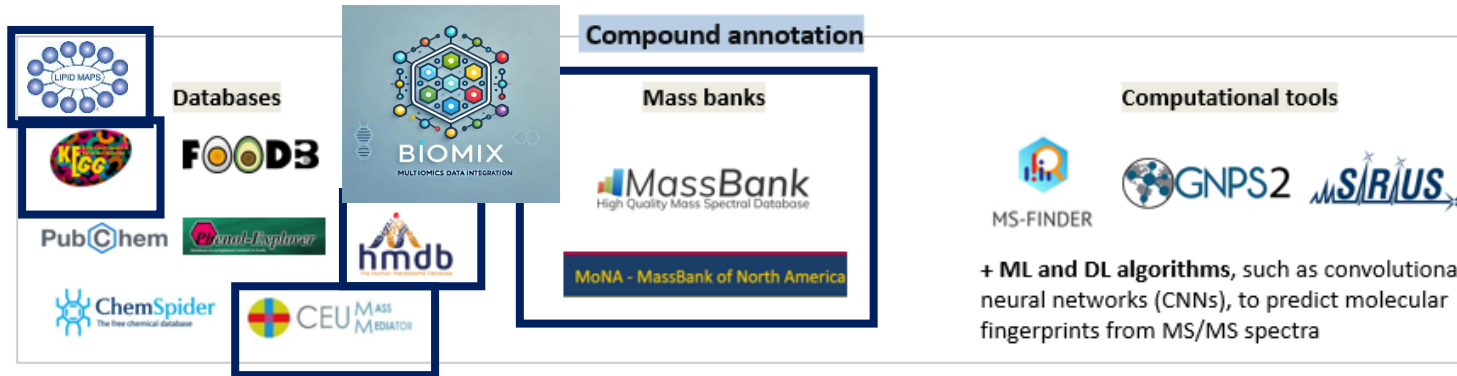


- Log2 fold change
- Pvalue
- Volcano Plot
- PCA



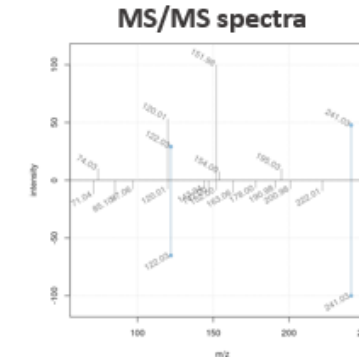
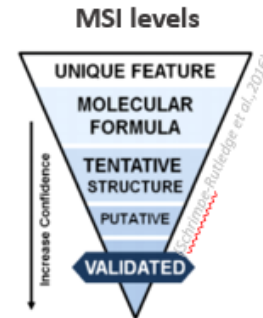


Metabolite annotation



Critical step (main bottleneck) in untargeted metabolomics

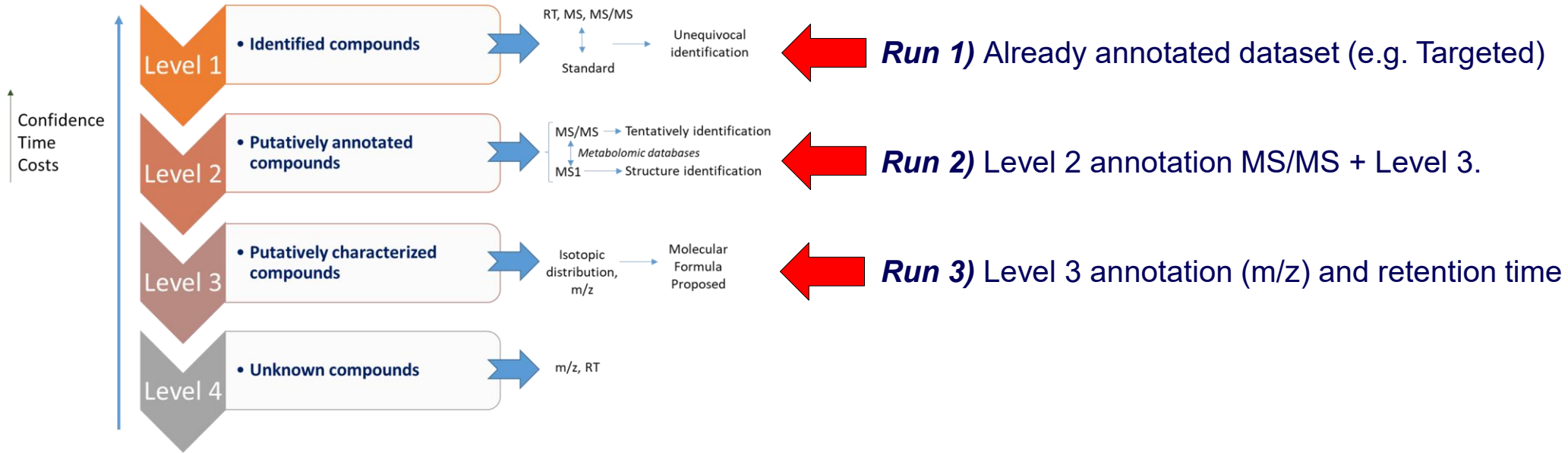
- Comparison of m/z , isotopic pattern, RT, and MS/MS or MSⁿ fragmentation spectra
- Report following the Metabolomics Standards Initiative (MSI) levels





Metabolomics analysis PTB

Same analysis different depth of annotation





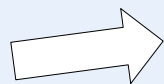
Metabolomics analysis PTB

⚙ Setup **Advanced Options** 📄 QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

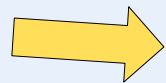
Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☒ MS1 only (non-targeted m/z matching against databases)
- ☐ MS1 + MS2 (m/z matching + spectral matching)

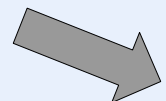


No annotation required (you have it already)

Targeted, or Untargeted with pre-annotated results



Level 3 annotation by (m/z) and retention time



Level 2 annotation by MS/MS + m/z and retention time

Untargeted metabolomics



1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file...

MTBLS7623_Metadata.xls

39 samples · 3 conditions: HC, PTB, PTB_DM

1) Load MTBLS7623_Metadata.xlsx from Metadata folder. (Metadata slot)

2. COMPARISON GROUPS

Condition (GROUP_1)

PTB

Control (GROUP_2)

HC

3. OUTPUT FOLDER

Folder where BiomiX will save results and the JSON configuration file

Select folder...

C:/Users/crist/Desktop/BiomiX2.5

2) Load MTBLS7623_positive_mode_annotated_HMDB.xls from Metabolomics folder. (Input 1)

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☒	Metabolomics	☐	TEST	☐	Browse file... MTBLS7623_positive_mode_annotated_HMDB.xls C:/Users/crist/Desktop/BiomiX2.5/Metabolomics/INPUT/MTBLS7623_positive_mode_annotated_HMDB.xls
Input 2	☐	☐	Undefined	☐	Label input 2	☐	Browse file... No file selected

	A	B	C	D
1	ID	HC_1	HC_10	HC_11
2	HMDB0012497	8958855	11731.87022	11096.62856
3	HMDB0000925	3062.01725	2260.162667	3635.334667
4	HMDB0034301	174354.5458	136631.2239	116859.7333
5	HMDB0034301	50937.20491	44507.57589	50626.45636
6	HMDB0031666	9584.321591	10557.13338	9649.629818
7	HMDB0000097	81410.99886	82769.48167	68992.30848



Metabolomics analysis PTB

⊙ Setup **Advanced Options** 📊 QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☒ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☐ MS1 + MS2 (m/z matching + spectral matching)

METABOLITE ANNOTATION

Metabolite annotation column

HMDB ▼

1) In Advance Options select the “Annotated mode”



Metabolomics analysis PTB

1) Open:
Metabolomics/OUTPUT/TEST_PTB_vs_HC

Nombre	
MetaboAnalyst	
Pathway_analysis	
plot_DMA_PTB_HC_TEST3	
PTB_HC_TEST3_metadata.tsv	
TEST3_PTB_HC_DOWN.tsv	
TEST3_PTB_HC_METABDOWN.tsv	
TEST3_PTB_HC_METABUP.tsv	
TEST3_PTB_HC_UP.tsv	
TEST3_PTB_vs_HC_peak_statistics.tsv	
TEST3_PTB_vs_HC_results.tsv	



Metabolomics analysis PTB

Nombre

- MetaboAnalyst
- Pathway_analysis
- plot_DMA_PTB_HC_TEST3
- PTB_HC_TEST3_metadata.tsv
- TEST3_PTB_HC_DOWN.tsv
- TEST3_PTB_HC_METABDOWN.tsv
- TEST3_PTB_HC_METABUP.tsv
- TEST3_PTB_HC_UP.tsv
- TEST3_PTB_vs_HC_peak_statistics.tsv
- TEST3_PTB_vs_HC_results.tsv

Only significant
negative hit

Only significant
positive hit

Results table

	A	B	C	D	E	F	G	H	I
1	shapiro_p	p_val	log2FC	NAME	padj	Name	chemical	average_m	kegg_id
2	0.082004	2.67E-05	-1.16687	HMDB0000	0.007518	Xanthine	C5H4N4O	152.1109	C00385
3	0.778672	0.000297	-0.66172	HMDB0003	0.039712	Racemeth	C5H11NO	149.211	C01733



Metabolomics analysis PTB

Nombre

MetaboAnalyst

Pathway_analysis

plot_DMA_PTB_HC_TEST3

PTB_HC_TEST3_metadata.tsv

TEST3_PTB_HC_DOWN.tsv

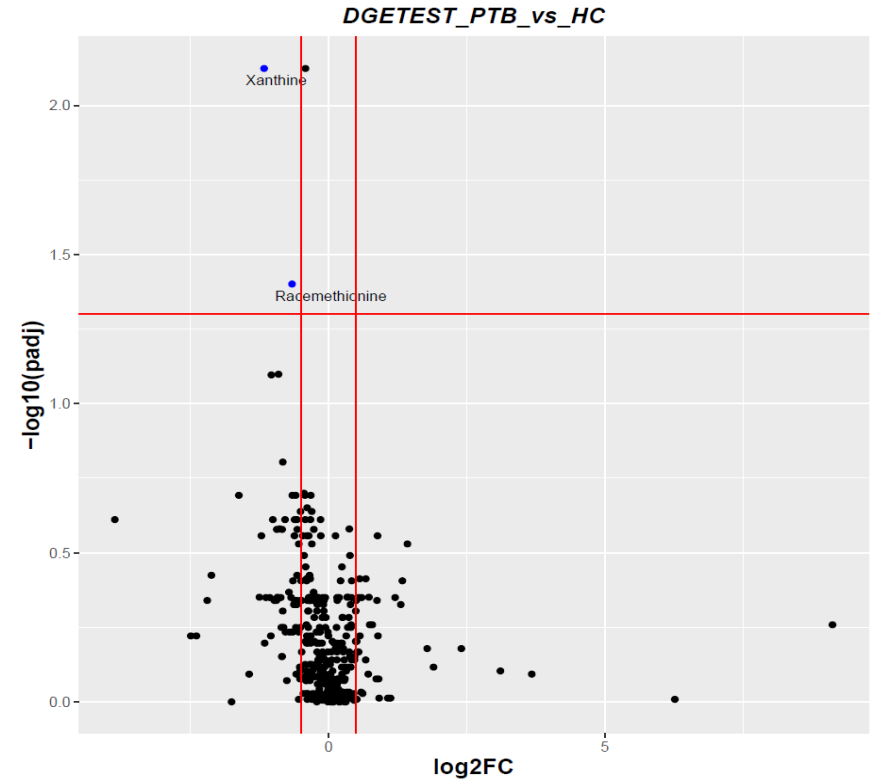
TEST3_PTB_HC_METABDOWN.tsv

TEST3_PTB_HC_METABUP.tsv

TEST3_PTB_HC_UP.tsv

TEST3_PTB_vs_HC_peak_statistics.tsv

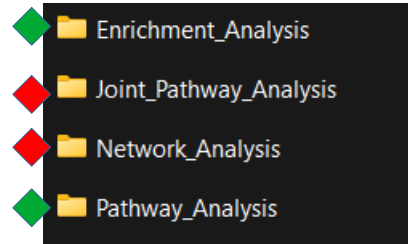
TEST3_PTB_vs_HC_results.tsv





- ◆ Combined with RNA output
- ◆ Only metabolomics output

Metabolomics analysis PTB



Nombre

MetaboAnalyst
Pathway_analysis
plot_DMA_PTB_HC_TEST3
PTB_HC_TEST3_metadata.tsv
TEST3_PTB_HC_DOWN.tsv
TEST3_PTB_HC_METABDOWN.tsv
TEST3_PTB_HC_METABUP.tsv
TEST3_PTB_HC_UP.tsv
TEST3_PTB_vs_HC_peak_statistics.tsv
TEST3_PTB_vs_HC_results.tsv



MetaboAnalyst 6.0

<https://www.metaboanalyst.ca/>



* To avoid to run it again with metabolomics include the transcriptomics input for metaboanalyst in the BiomiX 2.5

Example usage on Metaboanalyst

 **MetaboAnalyst 6.0** - from raw spectra to biomarkers, patterns, functions and systems biology

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Input Data Type	Available Modules (click on a module to proceed, or scroll down to explore a total of 18 modules including utilities)				
LC-MS Spectra (mzML, mzXML or mzData)			Spectra Processing [LC-MS w/wo MS2]		
MS Peaks (peak list or intensity table)		Peak Annotation [MS2-DDA/DIA]	Functional Analysis [LC-MS]	Functional Meta-analysis [LC-MS]	
Generic Format (.csv or .txt table files)	Statistical Analysis [one factor]	Statistical Analysis [metadata table]	Biomarker Analysis	Dose Response Analysis	Statistical Meta-analysis
Annotated Features		Enrichment Analysis	Pathway Analysis	Network Analysis	
Link to Genomics & Phenotypes (metabolite list)			Causal Analysis [Mendelian randomization]		

Enrichment_Analysis

Joint_Pathway_Analysis

Network_Analysis

Pathway_Analysis



* To avoid to run it again with metabolomics include the transcriptomics input for metaboanalyst in the BiomiX 2.5

Example usage on Metaboanalyst



MetaboAnalyst 6.0

- from raw spectra to biomarkers, patterns, functions and systems biology

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Input Data Type	Available Modules (click on a module to proceed, or scroll down to explore a total of 18 modules including utilities)				
LC-MS Spectra (mzML, mzXML or mzData)			Spectra Processing [LC-MS w/wo MS2]		
MS Peaks (peak list or intensity table)		Peak Annotation [MS2-DDA/DIA]	Functional Analysis [LC-MS]	Functional Meta-analysis [LC-MS]	
Generic Format (.csv or .txt table files)	Statistical Analysis [one factor]	Statistical Analysis [metadata table]	Biomarker Analysis	Dose Response Analysis	Statistical Meta-analysis
Annotated Features (metabolite list or table)		Enrichment Analysis	Pathway Analysis	Network Analysis	
Link to Genomics & Phenotypes (metabolite list)			Causal Analysis [Mendelian randomization]		



- Enrichment_Analysis
- Joint_Pathway_Analysis
- Network_Analysis
- Pathway_Analysis

- Transcriptomes_availables
- HMDB_ID_TEST_PTB_vs_HC.tsv

Copy paste

Please upload your data using one of the options below

Compound List

Concentration Table

Metabolomics Workbench Data

Metabolon Data Sheet

Joint Pathway Analysis

This module is designed for integrative analysis of transcriptomics/proteomics and metabolomics data [at pathway level](#).

- Metabolomics:**
 - Targeted metabolomics: a list of significant metabolites (or compounds of interest) with optional fold change values;
 - Untargeted metabolomics: a complete peak list. Peak names must be their numeric mass (m/z) values with optional retention times;
- Transcriptomics / Proteomics:**
 - A list of significant genes (or proteins) with optional fold change values;

Metabolomics Type:

Targeted (compound list) ▼

Compound list with optional fold changes

Copy paste

Genes/proteins with optional fold changes

ID Type:

--- Not Specified --- ▼

ID Type:

--- Not Specified --- ▼

> Submit

Try our example



Please upload your data using one of the options below

[Compound List](#)[Concentration Table](#)[Metabolomics Workbench Data](#)[Metabolon Data Sheet](#)[Joint Pathway Analysis](#)

This module is designed for integrative analysis of transcriptomics/proteomics and metabolomics data [at pathway level](#).

- **Metabolomics:**

- Targeted metabolomics: a list of significant metabolites (or compounds of interest) with optional fold change values;
- Untargeted metabolomics: a [complete](#) peak list. Peak names must be their numeric mass (m/z) values with optional retention times;

- **Transcriptomics / Proteomics:**

- A list of significant genes (or proteins) with optional fold change values;

Metabolomics Type:

Targeted (compound list) ▼

Compound list with optional fold changes

HMDB0000292	-1.166874014
HMDB0000159	-0.418125595
HMDB00033951	-0.66171578
HMDB0000673_5	-0.906404543
HMDB0000925	-1.035921633
HMDB0001999	-0.82879636
HMDB0011618	-0.444430066
HMDB0000929	-0.595395106
HMDB0010379	-1.623878436
HMDB0010381	-0.429302103
HMDB00008493	-0.322414705
HMDB00008138_11	-0.656334768
HMDB0000738	-0.388022968
HMDB0000267_2	-0.30533277
HMDB0000875	-0.507805485
HMDB0001921	-0.614559907
HMDB0000017	-0.6643663

ID Type:

HMDB ID ▼

Genes/proteins with optional fold changes

ZNF683	-2.302013132
CD69	-1.280647592
CTSW	-1.450083148
CCL4	-1.404571722
ID2	-0.719208523
FEZ1	-1.69089457
NKG7	-1.250631439
HNRNPDL	-0.607121002
FOSB	-3.053290289
PATL2	-0.709080192
G0S2	-2.305585463
CD3D	-0.933529732
NFU1	-0.63829913
XPNPEP2	-1.44088622
GZMM	-1.006252488
SPON2	-1.283466051
SLC15A3	-1.230430630

ID Type:

Official Gene Symbol ▼

> Submit

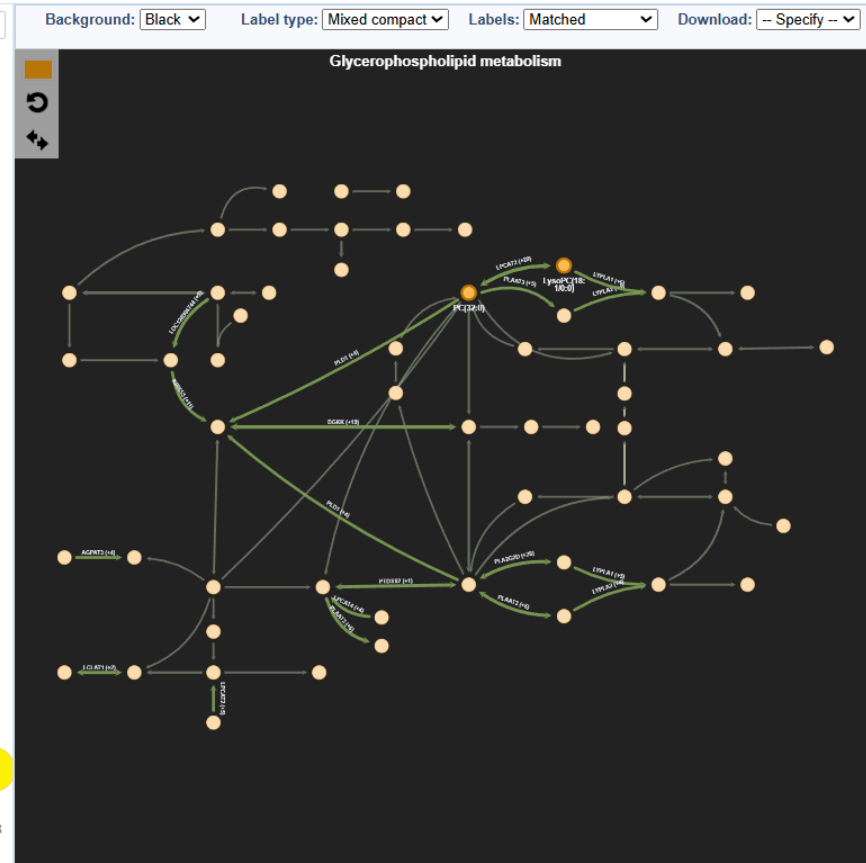
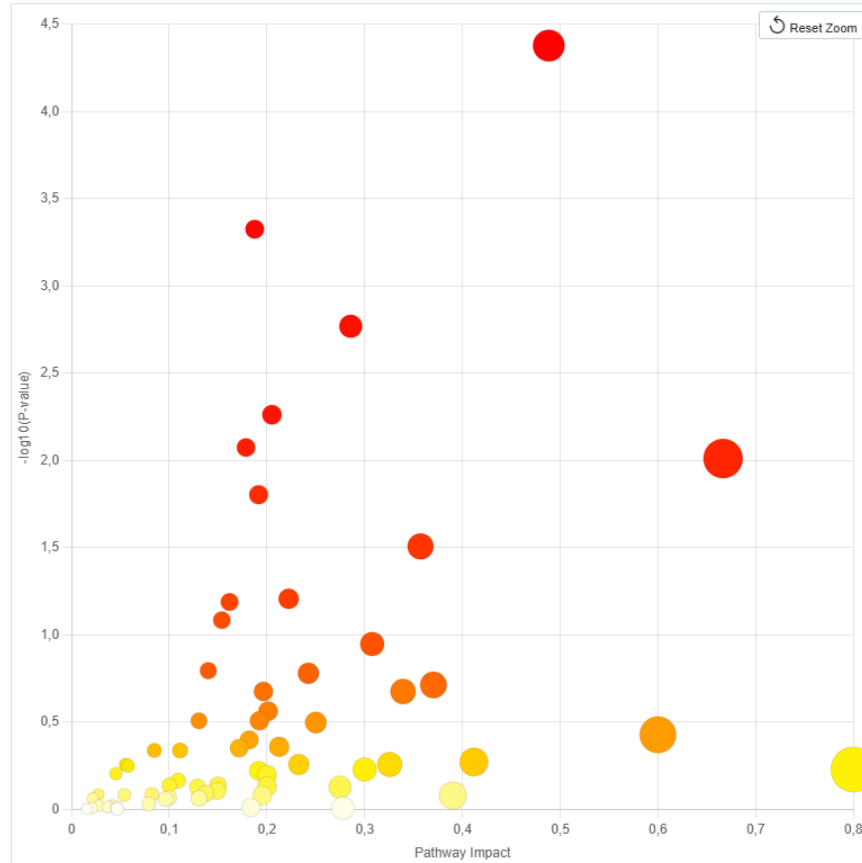
Try our example

Please upload your data using one of the options below

Show Pathway Analysis

Joint Pathway Analysis - Result View:

The scatter diagram summarizes matched pathways by pathway impact (x-axis) and enrichment significance (y-axis, $-\log_{10}$ p-value). Mouse over points to view pathway names and drag to zoom. Use reset to return to default view.



values;
tional retention times;

fold changes

002

nbol

> Submit

Try our example



How to annotate automatically?

2) Load MTBLS7623_positive_mode_unannotated.xls from Metabolomics folder. (Input 1)

OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☒	Metabolomics	☐	TEST	☐	Browse file... ✔ MTBLS7623_positive_mode_unannotated.xls /shared/Metabolomics/INPUT/MTBLS7623_positive_mode_u nannotated.xls
Input 2	☐	☐	Undefined	☐	Label input 2	☐	Browse file... No file selected



How to annotate automatically?

⊙ Setup **Advanced Options**  QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☒ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☐ MS1 + MS2 (m/z matching + spectral matching)

1) In Advance Options select the “MS1 only”

METABOLITE ANNOTATION

Metabolite annotation column

HMDB ▼



Positive or Negative mode?

Which adduct do you expect?

Which databases you want to consult to retrieve the annotation?

Setup Advanced Options QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diabolo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☒ MS1 only (non-targeted m/z matching against databases)
- ☐ MS1 + MS2 (m/z matching + spectral matching)

MS1 MATCHING PARAMETERS

Ion mode: positive M/Z tolerance (ppm): 15

MS1 ADDUCTS - POSITIVE MODE

☒ M+H	☒ M+2H	☒ M+Na	☒ M+K	☒ M+NH4	☒ M+H-H2O
---	--	--	---	---	---

MS1 ADDUCTS - NEGATIVE MODE

☐ M-H	☐ M-Cl	☐ M+FA-H	☐ M-H-H2O
------------------------------	-------------------------------	---------------------------------	----------------------------------

DATABASES MS1

☒ HMDB	☒ LipidMaps	☒ Metlin	☒ KEGG
--	---	--	--

MS1 ANNOTATION FILE (OPTIONAL)

Annotation file: Input index: No

Error match tolerance between
“your” peak and the database
match

The annotation should be
performed on which input
position?



3) Load
MTBLS7623_positive_mode_MS1_annotation.tsv
from Metabolomics folder. (Input 1)

	A	B	C
1	name	m/z	RT_min
2	peak1	70.06442	0.866483
3	peak2	76.07491	0.837358
4	peak3	86.09578	1.732158
5	peak4	86.09574	1.089042
6	peak5	91.05354	5.44075
7	peak6	104.1062	0.813258

SetupAdvanced OptionsQC & ImputationJSON Preview

GeneralMetabolomicsMOFA & DiabloSNF / NEMOMintTeaMetadata Filtering

Annotation mode

☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)

☒ MS1 only (non-targeted m/z matching against databases)

☐ MS1 + MS2 (m/z matching + spectral matching)

Ion mode

M/Z tolerance (ppm)

positive

15

MS1 MATCHING PARAMETERS

MS1 ADDUCTS - POSITIVE MODE

☒ M+H☒ M+2H☒ M+Na☒ M+K☒ M+NH4☒ M+H-H2O

MS1 ADDUCTS - NEGATIVE MODE

☐ M-H☐ M-Cl☐ M+FA-H☐ M-H-H2O

DATABASES MS1

☒ HMDB☒ LipidMaps☒ Metlin☒ KEGG

MS1 ANNOTATION FILE (OPTIONAL)

Annotation file

Input index

Browse...

No



-> Request of annotation to Ceu
Mass Mediator Server

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

+ Annotation_metabolites_B



TEST_PTb_vs_HC_results.tsv 29/06/2026 19:05 File TSV 49 KB

Adduct type

HMDB, KEGG and
PubChem ID and links

Statistics

Formula

Ppm error:
How much reliable?

Annotated name

m/z
Peak
number

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	m/z	NAME.x	shapiro_p	p_val	log2FC	RT_sec	RT_min	padj	retention	adduct_re	RT_score	identifier	name	formula	adduct	molecular	error_ppm	ionization	final_scor	KEGG	KEGG_UR	HMDB	HMDB_UR	LipidMaps	LipidMaps	Metlin
2	320.959	peak_114	3.614309	0.00013	#NOME?	60.265	1.00442	0.0674	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
3	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	117280	3-[3,5-dih	C14H100	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			
4	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	58543	3-hydroxy	C14H100	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			
5	452.953	peak_705	9.011311	0.00043	1.38237	69.2998	1.155	0.10778	1.155	-2	-2	69877	5-[3,5-dih	C14H100	M+Cl	417.984	1	-2	-2			HMDB012	http://www.hmdb.ca/metabolites/HMDB			



How to annotate automatically using fragmentation data (MS/MS)?

⊙ Setup **Advanced Options** 📊 QC & Imputation JSON Preview

General **Metabolomics** MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☒ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☐ MS1 + MS2 (m/z matching + spectral matching)

1) In Advance Options select the “MS1 + MS2”

METABOLITE ANNOTATION

Metabolite annotation column

HMDB ▼



Error match tolerance between
“your” peak and the database
match

Error match tolerance between “your”
peak m/z in MS1 and in MS2 data

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☒ MS1 + MS2 (m/z matching + spectral matching)

MS2 - ADDITIONAL PARAMETERS

MS2 RAW FILES DIRECTORY (.MZML)

M/Z tolerance MS2 (ppm) 25 RT match MS1-MS2 (sec) 10 Chromatography column rp Ion mode MS2 positive

MS2 ADDUCTS - POSITIVE MODE

☒ M+H ☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H-H2O

MS2 ADDUCTS - NEGATIVE MODE

☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

DATABASES MS2 (ASSIGN SEARCH PRIORITY)

HMDB 1 (priority) MONA 2 (priority) MassBank 3 (priority)

MS1 DATABASES USED IN MS2 WORKFLOW

☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 MATCHING IN MS2 WORKFLOW

MS1 tolerance (ppm) for MS2 15 MS2 annotation file index No MS2 annotation file (optional) Browse... Folder containing .mzML files Browse folder...



Public database priority in the MS2 match
(available: HMDB, MONA, Massbank)



Public database priority in the MS1 match
(available: HMDB, KEGG, Lipidmaps, Metlin)



Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☒ MS1 + MS2 (m/z matching + spectral matching)

MS2 - ADDITIONAL PARAMETERS

MS2 RAW FILES DIRECTORY (.mzML)

M/Z tolerance MS2 (ppm) 25 RT match MS1-MS2 (sec) 10 Chromatography column rp Ion mode MS2 positive

MS2 ADDUCTS - POSITIVE MODE

☒ M+H ☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H-H2O

MS2 ADDUCTS - NEGATIVE MODE

☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

DATABASES MS2 (ASSIGN SEARCH PRIORITY)

HMDB 1 (priority) MONA 2 (priority) MassBank 3 (priority)

MS1 DATABASES USED IN MS2 WORKFLOW

☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 MATCHING IN MS2 WORKFLOW

MS1 tolerance (ppm) for MS2 15 MS2 annotation file index No MS2 annotation file (optional) Browse... Folder containing .mzML files Browse folder...



3) Load
MTBLS7623_positive_mode_MS1_annotation.tsv
from Metabolomics folder. (Input 1)

	A	B	C
1	name	m/z	RT_min
2	peak1	70.06442	0.866483
3	peak2	76.07491	0.837358
4	peak3	86.09578	1.732158
5	peak4	86.09574	1.089042
6	peak5	91.05354	5.44075
7	peak6	104.1062	0.813258

SetupAdvanced OptionsQC & ImputationJSON Preview

GeneralMetabolomicsMOFA & DiablobNF / NEMOMintTeaMetadata Filtering

Annotation mode

☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)

☐ MS1 only (non-targeted m/z matching against databases)

☒ MS1 + MS2 (m/z matching + spectral matching)

MS2 - ADDITIONAL PARAMETERS

MS2 RAW FILES DIRECTORY (.M2ML)

M/Z tolerance MS2 (ppm)

RT match MS1-MS2 (sec)

Chromatography column

Ion mode MS2

MS2 ADDUCTS - POSITIVE MODE

☒ M+H

☒ M+2H

☒ M+Na

☒ M+K

☒ M+NH4

☒ M+H-H2O

MS2 ADDUCTS - NEGATIVE MODE

☐ M-H

☐ M-Cl

☐ M+FA-H

☐ M-H-H2O

DATABASES MS2 (ASSIGN SEARCH PRIORITY)

HMDB

MONA

MassBank

1 (priority)

2 (priority)

3 (priority)

MS1 DATABASES USED IN MS2 WORKFLOW

☒ HMDB

☒ LipidMaps

☒ Metlin

☒ KEGG

MS1 MATCHING IN MS2 WORKFLOW

MS1 tolerance (ppm) for MS2

MS2 annotation file index

MS2 annotation file (optional)

Folder containing .m2ml files

15

No

Browse...

Browse folder...



MS2_PLASMA1_1.mzML

MS2_URINE2_1.mzML

MS2_PLASMA_pool_MSMS-r001.mgf

pool_MSMS-r001.mgf

mgf_files

mzML_files

-> Provide the directory of the
MzML file or mgf file(s)

Name these file(s) including the
label used in the "label" i.e (TEST,
or PLASMA)

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diablo SNF / NEMO MintTea Metadata Filtering

Annotation mode

- ☐ Annotated (pre-annotated file with compound_name / HMDB / KEGG)
- ☐ MS1 only (non-targeted m/z matching against databases)
- ☒ MS1 + MS2 (m/z matching + spectral matching)

MS2 - ADDITIONAL PARAMETERS

MS2 RAW FILES DIRECTORY (.MZML)

M/Z tolerance MS2 (ppm) 25 RT match MS1-MS2 (sec) 10 Chromatography column rp Ion mode MS2 positive

MS2 ADDUCTS - POSITIVE MODE

☒ M+H ☒ M+2H ☒ M+Na ☒ M+K ☒ M+NH4 ☒ M+H-H2O

MS2 ADDUCTS - NEGATIVE MODE

☐ M-H ☐ M-Cl ☐ M+FA-H ☐ M-H-H2O

DATABASES MS2 (ASSIGN SEARCH PRIORITY)

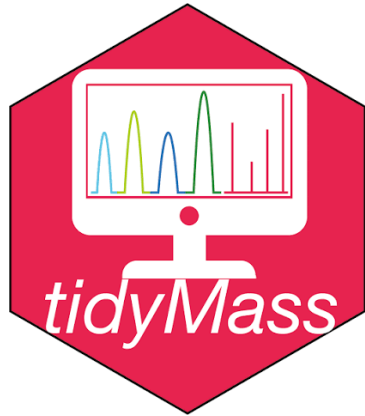
HMDB 1 (priority) MONA 2 (priority) MassBank 3 (priority)

MS1 DATABASES USED IN MS2 WORKFLOW

☒ HMDB ☒ LipidMaps ☒ Metlin ☒ KEGG

MS1 MATCHING IN MS2 WORKFLOW

MS1 tolerance (ppm) for MS2 15 MS2 annotation file index No MS2 annotation file (optional) Browse... Folder containing .mzML files Browse folder...



-> Request of annotation to Ceu
Mass Mediator Server

-> Local MS2 spectra extraction
and match on the databases
selected (locally availables).

MetaboAnalyst	18/03/2026 17:34	Cartella di file	
Pathway_analysis	29/06/2026 19:05	Cartella di file	
Heatmap_top_genes_TEST_HC_vs_PTB.pdf	18/03/2026 17:33	Adobe Acrobat D...	8 KB
plot_DMA_PTB_HC_TEST.pdf	29/06/2026 19:05	Adobe Acrobat D...	24 KB
PTB_HC_TEST_metadata.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_DOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABDOWN.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_METABUP.tsv	29/06/2026 19:05	File TSV	1 KB
TEST_PTB_HC_UP.tsv	29/06/2026 19:05	File TSV	0 KB
TEST_PTB_vs_HC_peak_statistics.tsv	29/06/2026 19:05	File TSV	27 KB
TEST_PTB_vs_HC_results.tsv	29/06/2026 19:05	File TSV	49 KB

+ Annotation_metabolites_B

+ MS2_SPECTRA

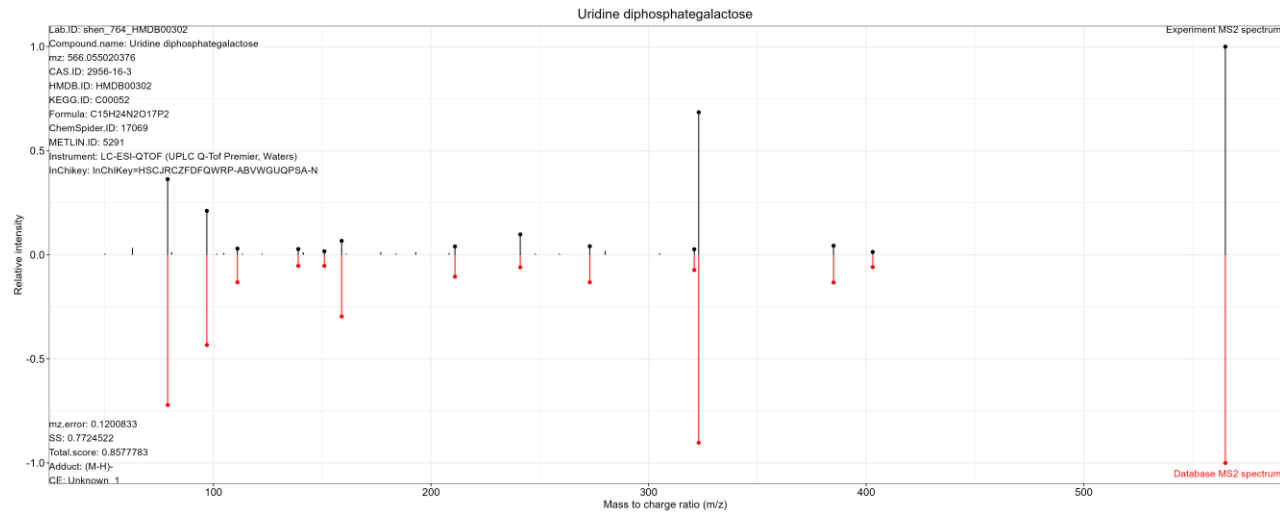


MS2_SPECTRA

ms2_match_plot_HMDB

ms2_match_plot_MASSBANK

ms2_match_plot_MONA



-> Results updated with MS2 annotation and folder including spectral match of the annotated peaks

HMDB_URI	LipidMaps	LipidMaps	Metlin	Metlin_URI	PubChem	PubChem	MS2_anno	annot_level
/dbget-bin/www_bget?cpd:C027			3596	https://me	11988267	https://pub	NA	Level_3
/dbget-bin/www_bget?cpd:C018			65671	https://me	25200469	https://pub	NA	Level_3
http://www.hmdb.ca/metabolite:			62907	https://me	439795	https://pub	NA	Level_3
http://www.hmdb.ca/metabolite:			3595	https://me	192	https://pub	NA	Level_3